

Autoinhibition of a clamp-loader ATPase revealed by deep mutagenesis and cryo-EM

Received: 10 July 2023

Accepted: 8 November 2023

Published online: 4 January 2024



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Clamp loaders are AAA+ ATPases that facilitate high-speed DNA replication. In eukaryotic and bacteriophage clamp loaders, ATP hydrolysis requires interactions between aspartate residues in one protomer, present in conserved 'DEAD-box' motifs, and arginine residues in adjacent protomers. We show that functional defects resulting from a DEAD-box mutation in the T4 bacteriophage clamp loader can be compensated by widely distributed single mutations in the ATPase domain. Using cryo-EM, we discovered an unsuspected inactive conformation of the clamp loader, in which DNA binding is blocked and the catalytic sites are disassembled. Mutations that restore function map to regions of conformational change upon activation, suggesting that these mutations may increase DNA affinity by altering the energetic balance between inactive and active states. Our results show that there are extensive opportunities for evolution to improve catalytic efficiency when an inactive intermediate is involved.

Clamp-loader complexes are oligomeric AAA+ ATPases that enable high-speed DNA replication by loading sliding clamps onto primer-template junctions for attachment to DNA polymerases^{1–4}. The subunits of AAA+ ATPase complexes consist of AAA+ modules that bind ATP at active sites located between subunits in the oligomer (Fig. 1a). Nature has elaborated on this conserved functional unit to generate a remarkable variety of molecular motors in which interfacial ATP binding and hydrolysis are coupled to a diversity of functional outputs. Members of different subfamilies of AAA+ ATPases are responsible for many processes in the cell, including DNA replication, the unwinding of DNA and RNA, the remodeling of protein complexes, and protein degradation^{5–11}. A fascinating question is how such diversity evolves while the function of the core catalytic engine that powers these motors is preserved.

AAA+ complexes form circular and spiral assemblies that are stabilized by interactions made by the ATP molecules that are bound at the interfaces between subunits. Nucleic acid or protein substrates are bound in a central channel formed by the spiral AAA+ assembly—in

the case of clamp loaders, this substrate is primer-template DNA, the binding of which triggers ATP hydrolysis by the clamp loader and the release of the closed clamp around the DNA^{1,12–15}. Some of the interfacial interactions in the active state have obvious importance for catalysis—for example, a highly conserved 'arginine finger' residue presented by one subunit coordinates the terminal phosphate of ATP bound to the other subunit^{16–18}. Other interfacial interactions are conserved within families, but not across all AAA+ ATPases, which raises the question of how the interacting residues change during evolution without loss of function.

An important set of family-specific interfacial interactions are formed by DEAD-box motifs found in a subset of AAA+ ATPases. In these AAA+ ATPases, including eukaryotic and bacteriophage clamp loaders, there is a consensus amino acid sequence motif, DEAD (hence why this motif is called the DEAD-box¹⁹), at the heart of the network of inter-subunit interactions. The first two residues of the DEAD-box motif (D and E) coordinate the magnesium ion in the ATPase active site and are

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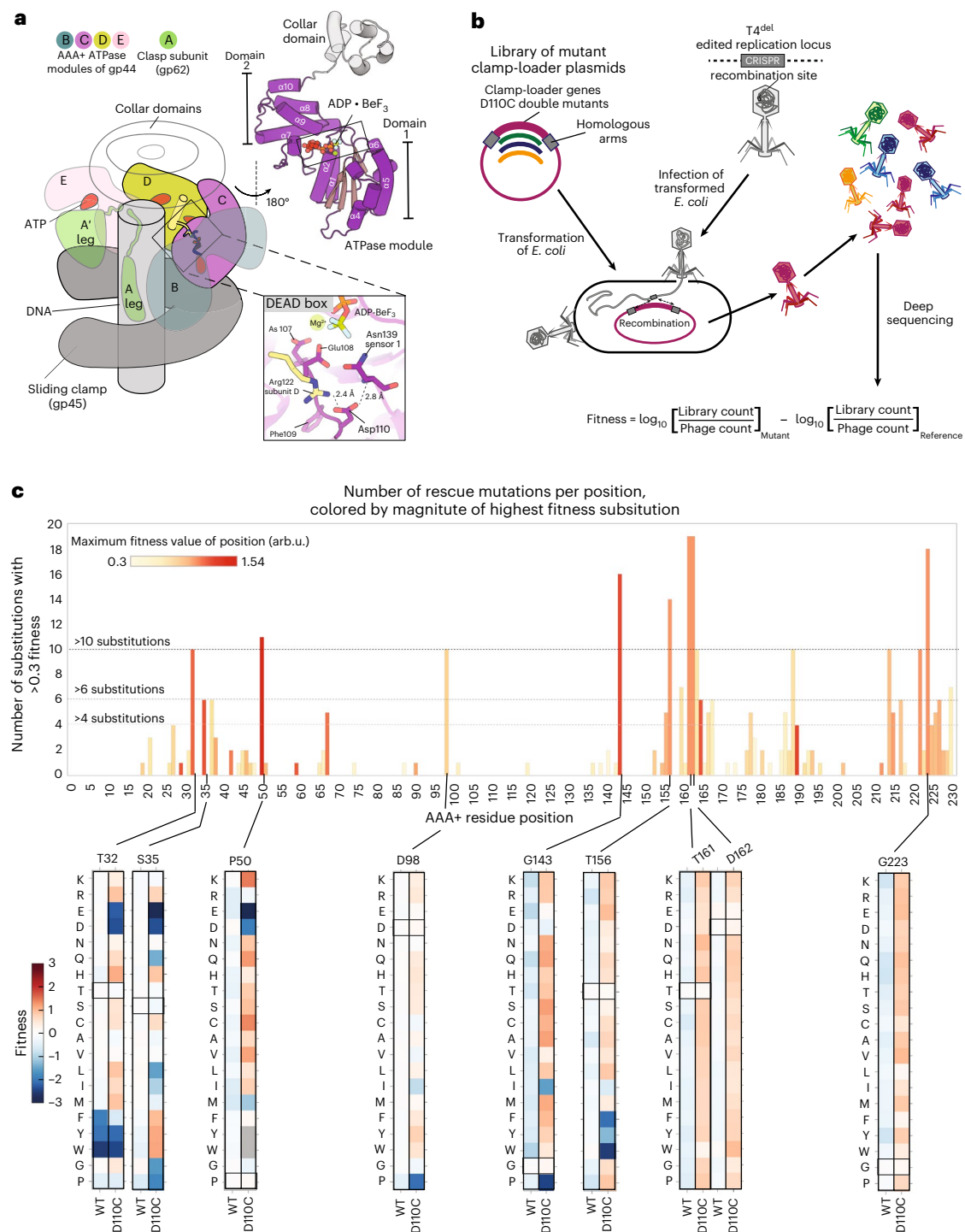


Fig. 1 | Clamp-loader structure and the T4 bacteriophage assay. a, The T4 clamp loader is bound to DNA and a sliding clamp (PDB 3U60). The five subunits of the clamp-loader complex are denoted A, B, C, D, and E. In the T4 complex, subunits B–E are the ATPase subunits (gp44) and are identical to each other. Subunit A (gp62) is degenerated and does not have an ATPase domain. We refer to subunit A as the ‘clasp’ subunit because it bridges the two open ends of the sliding clamp, using two extensions referred to as the A and A’ legs. The collar domains of each subunit form a pentameric assembly (transparent). Three catalytically competent ATP-binding sites are formed at the interfaces of subunits B–C, C–D, and D–E. Within each active site is a DEAD-box motif. Asp107 and Glu108 in the DEAD-box motif interact with the Mg²⁺ ion and nucleotide, and Asp110 coordinates Arg122 of a neighboring subunit. **b**, The platform uses a modified version of T4 bacteriophage (T4^{del}) in which the genes encoding the replication proteins of interest are deleted²⁰. Replication of T4^{del} is enabled by supplying the missing

replication proteins through helper plasmids. This is achieved by incorporating a Cas12–CRISPR system in the DNA encoding the missing replication proteins, which is recombined into the T4^{del} phage^{20,36}. The fitness value (also referred to as the ‘enrichment value’) is calculated by subtracting the quotient of the reference allele surviving selection from the selected mutant population. The plasmids that result in successful phage propagation can be counted by sequencing after cell lysis. **c**, Top, rescue mutations (fitness value > 0.3) are tallied (y axis) per residue position (x axis). The maximum fitness value per position that contains recovery mutations is noted by color on the corresponding mutation-count bar. Positions with a maximum fitness value of 0.3 are shown in light yellow, and positions that contain a recovery mutation and have a fitness value of 1.5 are shown in dark red. Bottom, nine rescue hotspots were identified, and these data were extracted for heatmaps in Fig. 2. The nine rescue hotspots allow for over six substitutions that recover mutant clamp loader fitness.

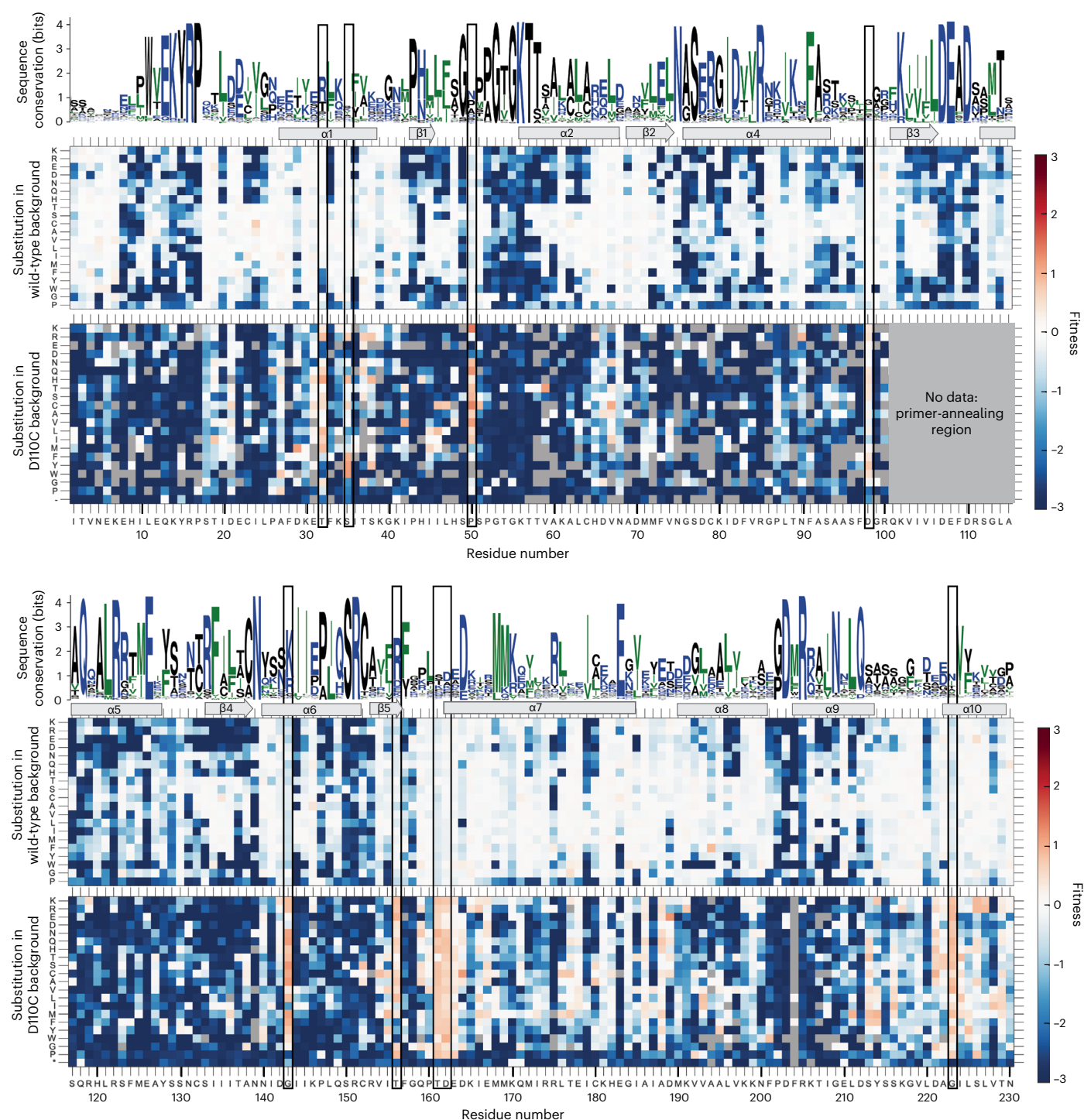


Fig. 2 | Deep mutagenesis of the p.D110C mutant clamp loader. The top panel in each block shows the sequence conservation, represented as a sequence logo³⁷, in the AAA+ modules of >1,000 clamp-loader sequences from phage genomic data²⁰. The first heat map in each block shows the mutational sensitivity of the wild-type T4 clamp loader, as determined previously²⁰. The second heatmap in each block shows the mutational sensitivity of the p.D110C variant, using D110C

as the reference allele. In these heatmaps, blue pixels correspond to variants that are less fit than the reference (white), with the minimum fitness (darkest blue) being 1,000-fold worse than that of the reference. Red pixels are variants that are more fit than the reference, whose maximum fitness is 1,000-fold better than that of the reference. Nine rescue hotspots are boxed in the two panels; these positions are discussed in the main text and are shown in Fig. 1c.

important for catalysis²⁰. The fourth residue in the DEAD box is typically an aspartic acid. In the clamp loaders it forms an ion-pairing interaction with an arginine side chain that is presented by the adjacent AAA+ subunit. This interfacial ion pair, anchored by the DEAD-box motif, is positioned at a key juncture of the interactions that link DNA recognition to the formation of catalytically competent active sites (Fig. 1a).

We were struck by the fact that the DEAD-box motif, which is so centrally important to the integrity of the active states of the eukaryotic and bacteriophage clamp loaders^{15,21,22}, is incomplete or absent in other AAA+ ATPases, including the bacterial clamp loaders¹². To better understand the role of the DEAD-box motif in clamp-loader function, we studied the T4 bacteriophage clamp loader, a well-established model

system^{23,24} for which we have developed a high-throughput mutagenesis platform (see Fig. 1a for a schematic of the clamp loader and the notation used to describe subunits; the T4 clamp loader consists of one gp62 subunit, denoted A, and four gp44 ATPase subunits, denoted B, C, D, and E). We mapped the mutational responses of a T4 clamp-loader variant with a mild defect in DNA-dependent ATP hydrolysis. This defect was created by replacing the interfacial aspartate residue in the DEAD-box motif of the ATPase subunit (Asp110 in the gp44 ATPase subunit of the T4 clamp loader) with cysteine (p.D110C), which attenuates the DNA-dependent ATPase rate of the clamp loader. The relatively mild phenotype of this alteration²⁰ facilitates the identification of second-site substitutions that rescue the fitness of the p.D110C mutant clamp loader. Such rescue mutations can potentially provide insights into the role of the DEAD-box motif in the clamp-loader mechanism.

We carried out saturation mutagenesis of the p.D110C variant of the T4 clamp loader and found that substitutions at many positions in the AAA+ module rescue fitness, as measured by phage propagation. Substitutions at these sites do not affect the fitness of the wild-type clamp loader²⁰. Interestingly, in the p.D110C background, the rescue function could be fulfilled by different amino acid substitutions at the same site. We interpret the presence of such rescue hotspots as evidence for an unidentified inactive state of the T4 clamp loader—in this interpretation, the mutations act by destabilizing this inactive state, and the precise identity of the substitution is less important. It is important to note that the rescue alterations could restore activity in other ways, such as by affecting clamp affinity or ATPase activity, although it is more difficult to explain the diversity of activating substitutions in this way. We used cryogenic electron microscopy (cryo-EM) to identify and structurally characterize such an inactive state and found that it is a stable conformation of the clamp-loader complex in the absence of DNA. Our results suggest that p.D110C increases the affinity of the clamp loader for DNA by altering the energetic balance between the active DNA-bound state and the inactive state to which DNA cannot bind.

Results

Identification of DEAD-box rescue mutations

We have recently developed a platform for high-throughput mutagenesis of T4 bacteriophage replication proteins in their proper functional context²⁰ (Fig. 1b and Methods). A library of helper plasmids is created, each bearing a different variant of the relevant replication proteins. The functional fitness is quantified by using deep sequencing to count the variants in the input helper plasmid library and also in the phage that are released after lysis of the *Escherichia coli* cells. The fitness of a variant is defined as:

$$\text{fitness} = \log_{10} \left[\frac{\text{library count}}{\text{phage count}} \right]_{\text{mutant}} - \log_{10} \left[\frac{\text{library count}}{\text{phage count}} \right]_{\text{reference}}$$

Here, ‘mutant’ refers to the particular mutation for which the fitness is being measured, and ‘reference’ is the allele (for example, wild-type or D110C) that serves as the reference. In this logarithmic fitness scale, a value of +1 corresponds to a tenfold faster rate of phage propagation than that of the reference variant, and a fitness score of −1 corresponds to a tenfold slower rate of propagation.

We determined, using the phage propagation assay, that p.D110C leads to a reduction of phage propagation of approximately sixfold (Extended Data Fig. 1a). Other mutations, such as that encoding D110S, reduce phage propagation by a factor of ~100 or more. Biochemical measurements with the purified wild-type T4 clamp loader showed that it has an affinity for primed DNA in the nanomolar range (dissociation constant (K_d), approximately 40 nM) in the presence of the sliding clamp (Extended Data Fig. 2a). Under identical conditions, the p.D110C clamp loader mutant showed no detectable DNA binding. The p.D110C clamp loader does, however, retain a weak affinity for DNA,

Table 1 | Data collection and refinement statistics

	p.D110C T4 bacteriophage clamp loader with sliding clamp and DNA (8UK9)	Wild-type T4 bacteriophage clamp loader with sliding clamp and DNA (8UH7)
Data collection ^a		
Space group	<i>P</i> 2 ₁ 2 ₁ 2 ₁	<i>P</i> 121
Cell dimensions		
<i>a</i> , <i>b</i> , <i>c</i> (Å)	95.243, 231.991, 264.511	93.35, 118.32, 132.89
<i>a</i> , <i>b</i> , <i>c</i> (°)	90, 90, 90	90.00, 102.05, 90.00
Resolution (Å)	46.87–3.1	49.65–2.63
	(3.211–3.1) ^b	(2.92–2.63)
<i>R</i> _{merge}	0.05745 (1.571)	0.325 (2.096)
<i>I</i> /σ(<i>I</i>)	5.80 (0.46)	6.9 (1.1)
Completeness (%)	Ellipsoidal, 95.1 (64.0)	Ellipsoidal, 92.9 (68.3)
Redundancy	2.0 (2.0)	6.3 (6.9)
Refinement		
Resolution (Å)	46.87–3.1	49.65–2.64
	(3.211–3.1)	(2.73–2.64)
No. reflections	76,439 (470)	55,477 (250)
<i>R</i> _{work} / <i>R</i> _{free}	0.2525 / 0.2643	0.1948 / 0.2466
No. atoms		
Protein	35,422	17,800
Ligand/ion	248	124
Water	0	34
<i>B</i> factors		
Protein	131.1	46.49
Ligand/ion	94.67	26.04
Water	0	26.00
R.m.s. deviations		
Bond lengths (Å)	0.003	0.003
Bond angles (°)	0.66	0.54

^aOne crystal was used for each structure. ^bValues in parentheses are for highest-resolution shell.

because the addition of DNA stimulates the rate of ATP hydrolysis (Extended Data Fig. 2b). We determined the crystal structure of the p.D110C clamp-loader variant bound to primed DNA and the sliding clamp, relying on high protein and DNA concentration to drive formation of the complex (Extended Data Figs. 1b and 3 and Table 1). The overall structure of the complex is very similar to that of the wild-type complex²², and the structure suggests that Cys110 is deprotonated and negatively charged at those sites, thereby mimicking the aspartate residue and explaining the mild phenotype of the mutant. There are two complexes in the asymmetric unit of the crystal. At four of the six interfacial ATPase sites in the two complexes, the termini of the cysteine side chains are within approximately 3 Å of the termini of the side chains of Arg122, from adjacent subunits, consistent with ion-pair formation.

For saturation mutagenesis of the p.D110C clamp loader, we used a library for the AAA+ module of the T4 clamp loader that had been developed previously²⁰, and we fixed the D110C point mutation into this library (Methods). The resulting double-mutant library (the D110C mutation plus additional single-site mutations) was used for phage propagation. The D110C phage DNA sequence without a second mutation is used as the reference allele when calculating fitness (see Fig. 1b). In this logarithmic fitness scale, the p.D110C clamp loader itself has a fitness value of zero. Extended Data Fig. 4 shows the reproducibility of these fitness scores.

The p.D110C variant of the clamp loader is markedly more sensitive to mutations than is the wild-type complex. Strikingly, the saturation-mutagenesis data identify mutations that restore the fitness of the p.D110C variant (regions shaded red in Figs. 1c and 2). For this analysis, we considered mutations that yield a fitness value of >0.3, corresponding to a twofold increase of phage propagation over the variant with the p.D110C clamp loader, as gain-of-fitness variants (‘rescue mutations’). Using this criterion, we found that approximately 80 positions in the AAA+ module have at least one rescue mutation (see red boxes and bar plot in Fig. 1c). Of these positions, ten have at least one mutation with a fitness score above 1.0, corresponding to wild-type levels of activity, or essentially complete repair of the replication defect. Thus, the clamp loader has a distributed capacity to compensate for the defect in the DEAD-box motif through point mutations.

Purification and biochemical measurements of clamp-loader complexes with five rescue mutations show that these mutations increase DNA affinity, the ATP hydrolysis rate, or both relative to those of the p.D110C clamp loader (Extended Data Fig. 5). For example, the p.P50K variant of the p.D110C clamp loader has DNA-binding affinity and DNA-stimulated ATPase rates that are both comparable to those of the wild-type clamp loader. The DNA affinity of the p.G143R variant of the p.D110C clamp loader is comparable to that of wild-type clamp loader, but in this case the ATPase rate is not completely recovered. Other mutations have weaker effects. The D98W mutation has a small increase in DNA affinity, but no improvement in ATP hydrolysis.

The locations of the rescue mutations do not readily explain why these mutations improve fitness. The rescue mutations do not occur at sites of obvious functional importance, because they do not overlap with sites with conserved sequences (Fig. 2 and Extended Data Fig. 6). A clue to the mechanism of rescue comes from observing an unexpected pattern in the deep mutagenesis data. At many sites where rescue mutations occur, more than one kind of mutation results in improved fitness. We refer to such sites as ‘rescue hotspots.’ Nine of the most prominent rescue hotspots are highlighted in Figs. 1c and 2. Some of these rescue hotspots are sites at which ten or more substitutions increase the fitness of the p.D110C variant. The average value of the maximum fitness score for substitutions at these sites is approximately 1.1, which corresponds to wild-type levels of fitness. The ability of very different substitutions at a particular site to rescue fitness suggests that these mutations may increase the activity of the p.D110C clamp loader by non-specific effects, which we demonstrate to be the destabilization a less active autoinhibited state, but other factors involving, for example, changes in clamp binding, DNA binding, or ATPase activity, could also be involved.

An inactive conformation of the clamp loader without DNA

We used cryo-EM to determine the structure of the wild-type T4 clamp-loader complex in the presence and absence of primed DNA and the sliding clamp (Table 2 and Extended Data Fig. 7). In order to minimize perturbation of the equilibrium between different conformational states of the clamp loader complex, we did not use cross-linking reagents in any of our cryo-EM samples. We first determined the cryo-EM structure of the clamp loader–clamp complex in the presence of primed DNA and ADP-BeF₃. The resulting structure closely resembles a previously determined crystal structure of the same complex²² (Fig. 3a). If the two structures are aligned using the AAA+ module of subunit C, the r.m.s. deviation in Cα positions calculated over the other AAA+ modules (residues 1–230 of chains B, D, and E) is approximately 2.4 Å. The ATP analog, ADP-BeF₃, is visualized at the ATPase sites of the B, C, and D subunits, with the BeF₃ moiety at each site coordinated by the sidechain of the arginine-finger residue (Arg151) from neighboring subunits. Cryo-EM density corresponding to an ADP molecule is seen at the ATPase site of subunit E, consistent with the crystal structure²². We refer to this structure as the ‘active complex.’

Table 2 | Cryo-EM data collection, refinement and validation statistics

	#1 DNA-bound clamp loader–clamp (EMD-42399) (PDB 8UNF)	#2 DNA-free clamp loader–clamp (EMD-42402) (PDB 8UNH)
Data collection and processing		
Magnification	×81,000	×81,000
Voltage (kV)	300	300
Electron exposure (e [−] /Å ²)	50	50
Defocus range (μm)	−0.8 to −2.2	−0.8 to −2.2
Pixel size (Å)	0.524	0.524
Symmetry imposed	No	No
Initial particle images (no.)	4,206,415	17,937,844
Final particle images (no.)	204,448	455,041
Map resolution (Å)	3.15	3.21
FSC threshold	0.143	0.143
Refinement		
Initial model used (PDB code)	3U60	3U60, 3U61
Map sharpening B factor (Å ²)	−133	−157
Model composition		
Non-hydrogen atoms	17,810	13,271
Protein residues	2,147	1,683
Ligands	8	4
B factors (Å²)		
Protein	64.78	50.59
Ligand	47.45	58.12
R.m.s. deviations		
Bond lengths (Å)	0.007	0.004
Bond angles (°)	0.662	0.480
Validation		
MolProbity score	1.92	2.52
Clashscore	11.50	46.50
Rotamer outliers(%)	0.44	0.49
Cb outliers (%)	0.05	0.00
Ramachandran plot		
Favored (%)	95.12	94.48
Allowed (%)	4.69	5.28
Disallowed (%)	0.19	0.24

We did not observe the formation of this DNA-bound active complex of the wild-type clamp loader if the p.D110C variant of the clamp loaders was used during sample preparation. We also did not observe the formation of the active complex if the non-hydrolysable ATP analog ATPγS was used instead of ADP-BeF₃. Using the fluorescence anisotropy assay described above, we confirmed that DNA binding to the wild-type T4 clamp-loader complex could be detected readily in the presence of ADP-BeF₃ (K_d, approximately 40 nM) or ADP-AlF₃ (K_d, approximately 100 nM), but not with ATPγS or ADP alone, or in the absence of nucleotides (Fig. 4a). The inability of the T4 clamp loader to bind DNA in the presence of ATPγS^{25,26}, along with the results of saturation mutagenesis on the p.D110C variant that showed numerous gain of function sites, motivated us to pursue structural analysis of a DNA-free form of the clamp loader.

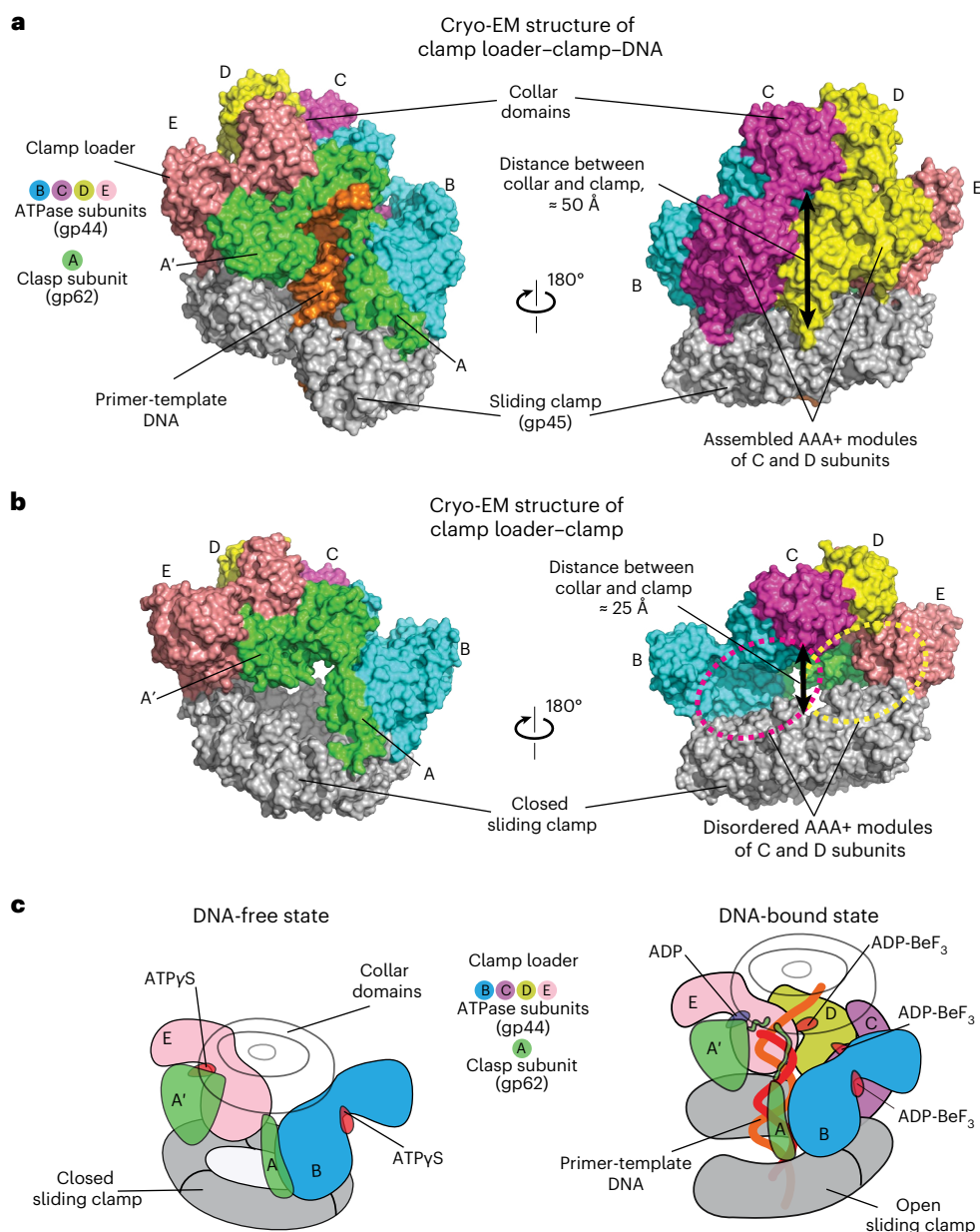


Fig. 3 | Cryo-EM structures of DNA-bound and DNA-free state of the clamp-loader complex. **a**, The structure of the T4 clamp loader is shown, including subunits A (gp62, green), B (gp44, cyan), C (gp44, magenta), D (gp44, yellow), and E (gp44, salmon). The open T4 sliding clamp (gp45) is colored gray, and primer-template DNA is colored orange. The clasp subunit (gp62, green) bridges the gap in the open sliding clamp with its A leg on the lower part of the clamp and its A' leg at the top. The primer-template DNA (orange) is bound in the central chamber of the clamp loader–clamp complex, with the single-stranded template overhang threaded through the gap between the A and A' legs. **b**, Surface representation of the cryo-EM structure of the clamp loader–clamp complex in the absence of DNA (same color scheme as in **a**). The T4 sliding clamp (gp45, gray) is closed. The AAA+ modules of subunits

C and D are disordered in this structure (right, magenta and yellow). The distance between the collar domains of subunits C and D and the sliding clamp reduces from approximately 50 Å in the DNA-bound complex (**a**, right) to approximately 25 Å in the DNA-free complex (**b**, right), as indicated by the length of the double-headed black arrows. **c**, Schematics (colored as in **a**) of the T4 clamp loader–clamp complex in the DNA-free (left) and DNA-bound (right) states. For clarity, the collar domains are not colored. ATPγS molecules (red) are bound to subunits B and E of the clamp loader in the DNA-free state. In the DNA-bound state of the clamp loader, ADP-BeF₃ is bound to subunits B, C, and D, and ADP (purple) is bound to subunit E.

We used cryo-EM to analyze the p.D110C variant of the clamp loader–clamp complex in the presence of ADP-BeF₃, and the wild-type clamp loader–clamp complex in the presence of ATPγS. In both of these conditions, the clamp loader adopted a new DNA-free conformation, even when primed DNA was used during sample preparation and was present on the grids (Fig. 4c). The cryo-EM structure of the wild-type T4 clamp loader that was bound to ATPγS and the sliding clamp but had no bound DNA was determined at a resolution of 3.2 Å

(Fig. 3b). The collar domains form a pentameric assembly, as in other clamp-loader structures. The collar assembly packs more closely against the AAA+ modules of subunits B and E than is the case for the DNA-bound complex. Only the AAA+ modules of subunits B and E were resolved in this cryo-EM structure. Additional diffuse density can be identified in the vicinity of the presumed locations of the AAA+ modules of subunits C and D, at lower contour levels of the cryo-EM map. We used different data-processing strategies, including partial

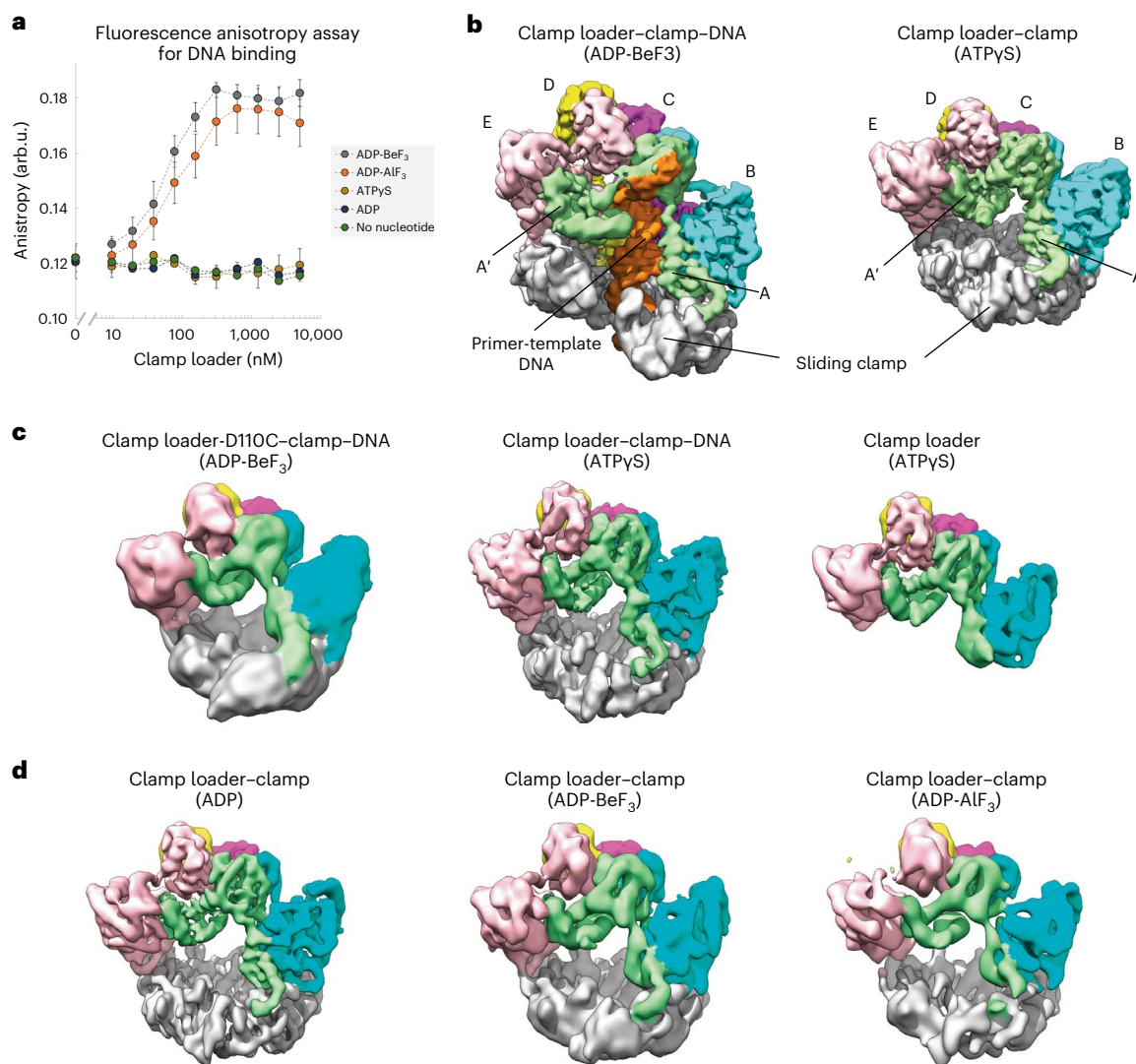


Fig. 4 | The clamp loader adopts a stable inactive conformation. **a**, The fluorescence anisotropy assay was used to measure DNA binding to the clamp loader-clamp complex in the presence of 1 mM ADP, ATPγS, ADP-BeF₃, ADP-AlF₃, or no nucleotide. In this assay, DNA binding to the clamp loader-clamp complex is detectable only in the presence of ADP-BeF₃ or ADP-AlF₃. Data are presented as mean values $\pm$ s.d.; $n = 3$ biologically independent experiments. **b**, Cryo-EM maps of the wild-type clamp loader-clamp-DNA complex in the presence of ADP-BeF₃ (left) and the wild-type clamp loader-clamp complex in the presence of ATPγS (right) are colored as in Fig. 3. **c**, Cryo-EM maps of the clamp loader-D110C-clamp complex in the presence of ADP-BeF₃ (left) and

the wild-type clamp loader-clamp complex in the presence of ATPγS (middle) show that both of these two clamp-loader complexes adopt the DNA-free conformation, even when primed DNA is used during sample preparation and is present on the EM grids. Right, the clamp loader alone in the presence of ATPγS adopts the same DNA-free conformation. **d**, Cryo-EM maps of the wild-type clamp loader-clamp complex in the presence of ADP (left), ADP-BeF₃ (middle), and ADP-AlF₃ (right) show that these complexes all adopt the distorted DNA-free conformation described in a, independent of the nature of the nucleotide bound to the ATPase sites.

signal subtraction, masked three-dimensional (3D) classification, and masked 3D refinement, but we were unable to resolve interpretable density for the AAA+ modules of subunits C and D, which implies they are mobile and are not held in a fixed position by their neighboring subunits. We refer to this complex as the ‘inactive complex,’ because the interfacial catalytic sites are not assembled.

We also used cryo-EM to analyze the T4 clamp loader-clamp complex without DNA, in the presence of ADP, ADP-BeF₃, or ADP-AlF₃. The clamp-loader complex adopts the same distorted conformation seen in the ATPγS-bound complex under all of these conditions (Fig. 4d), suggesting that this DNA-free conformation of the T4 clamp loader is stable and is insensitive to the nature of the nucleotide bound to the ATPase domain. We also used cryo-EM to analyze the clamp loader in the absence of the sliding clamp and DNA. The clamp loader alone adopts the same DNA-free inactive conformation (Fig. 4c).

Inactive conformation of the clamp loader blocks DNA binding

The structures of the DNA-bound and DNA-free states of the clamp loader-clamp complex provide an explanation for why DNA binding is required to form the catalytically active complex. In a DNA-bound clamp loader, the single-stranded portion of the primed DNA binds to the collar domain of the clasp subunit (subunit A), between the two linkers connecting the collar domain to the A and the A' legs. In the absence of DNA, these linkers switch their conformations, from extended loops in the active conformation to α -helices in the inactive conformation (Fig. 5a). This conformational switching leads to a tighter interaction between the collar domain of the clasp subunit and the AAA+ module of the B subunit in the inactive conformation. This reduces the distance between the collar domains of subunits C and D and the sliding clamp, from approximately 50 Å in the DNA-bound complex to approximately 25 Å in the DNA-free state (Fig. 3a,b).

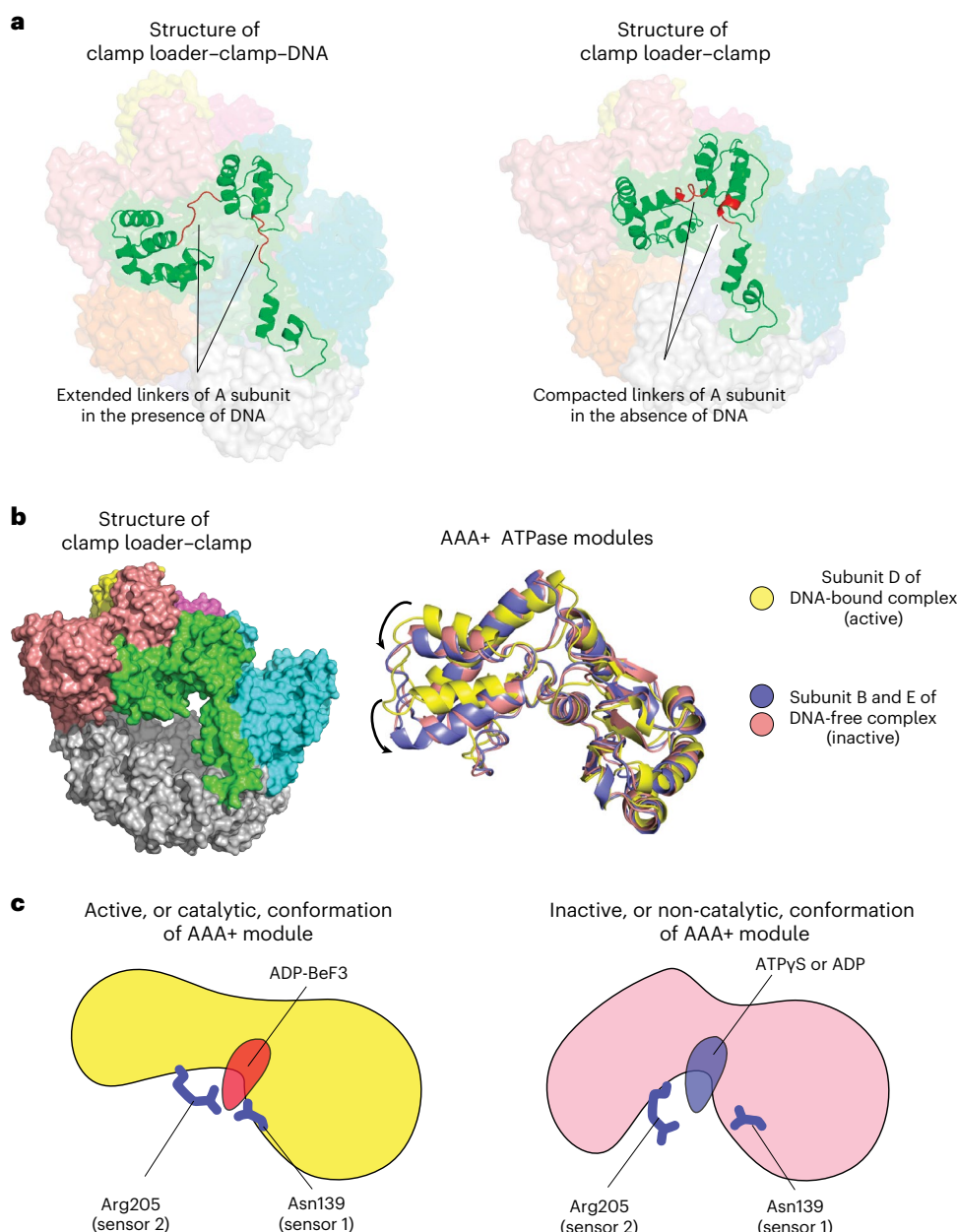


Fig. 5 | Conformational differences between the active and inactive states.

a, Surface representations of the cryo-EM structures of the DNA-bound (left) and the DNA-free (right) states of the clamp-loader complex are superimposed with cartoon representations of the clasp subunit (gp62, green). The linker regions (red) connecting the collar domain to the A and the A' legs in the clasp subunit switch their conformations, from extended loops (left, DNA-bound state) to α -helices (right, DNA-free state). This conformational switch allows the collar domains to pack more closely to the AAA+ module of subunit B, which reduces the distance between the collar domains and the sliding clamp (Fig. 4a,b). **b**, AAA+ modules of subunits B (purple) and E (salmon) of the inactive DNA-free state of the clamp-loader complex are superimposed with subunit D (yellow) of the active DNA-bound state of the clamp-loader complex. These structures are aligned using domain 1 (residues 1–120) of the AAA+ module. Black arrows highlight the inwards rotation (of approximately

10°) of domain 2 (residue 160–230) towards domain 1, observed in subunits B and E of the DNA-free complex when compared with subunit D of the DNA-bound complex. **c**, Schematics of the active (left, yellow) and inactive (right, pink) conformations of the AAA+ modules. The active conformation of the AAA+ module is adopted by subunits B, C, and D of the DNA-bound state of the clamp-loader complex. The inactive conformation of the AAA+ module is adopted by subunits B and E of the DNA-free state and subunit E of the DNA-bound state of the clamp-loader complexes. In the active conformation, key residues in the ATPase sites, Asn139 (sensor 1) and Arg205 (sensor 2), are positioned near ADP–BeF₃ in a manner that is consistent with the formation of a catalytically competent active site. In the inactive conformation, these two residues are pulled away from the position of the terminal phosphate of ATPγS or ADP.

The compressed space between the collar domain and the sliding clamp can no longer accommodate DNA or the AAA+ modules of the C and D subunits, which are displaced from the main body of the assembly. As a result, DNA cannot bind, and none of the catalytic ATPase sites between the AAA+ modules are assembled.

Rescue hotspots located near sites of conformational change Comparison of the cryo-EM structures shows that rescue hotspots for p.D110C are in regions of the AAA+ module that undergo structural change as the T4 clamp loader transitions between the inactive DNA-free and active DNA-bound states (Fig. 6a,b). To analyze these

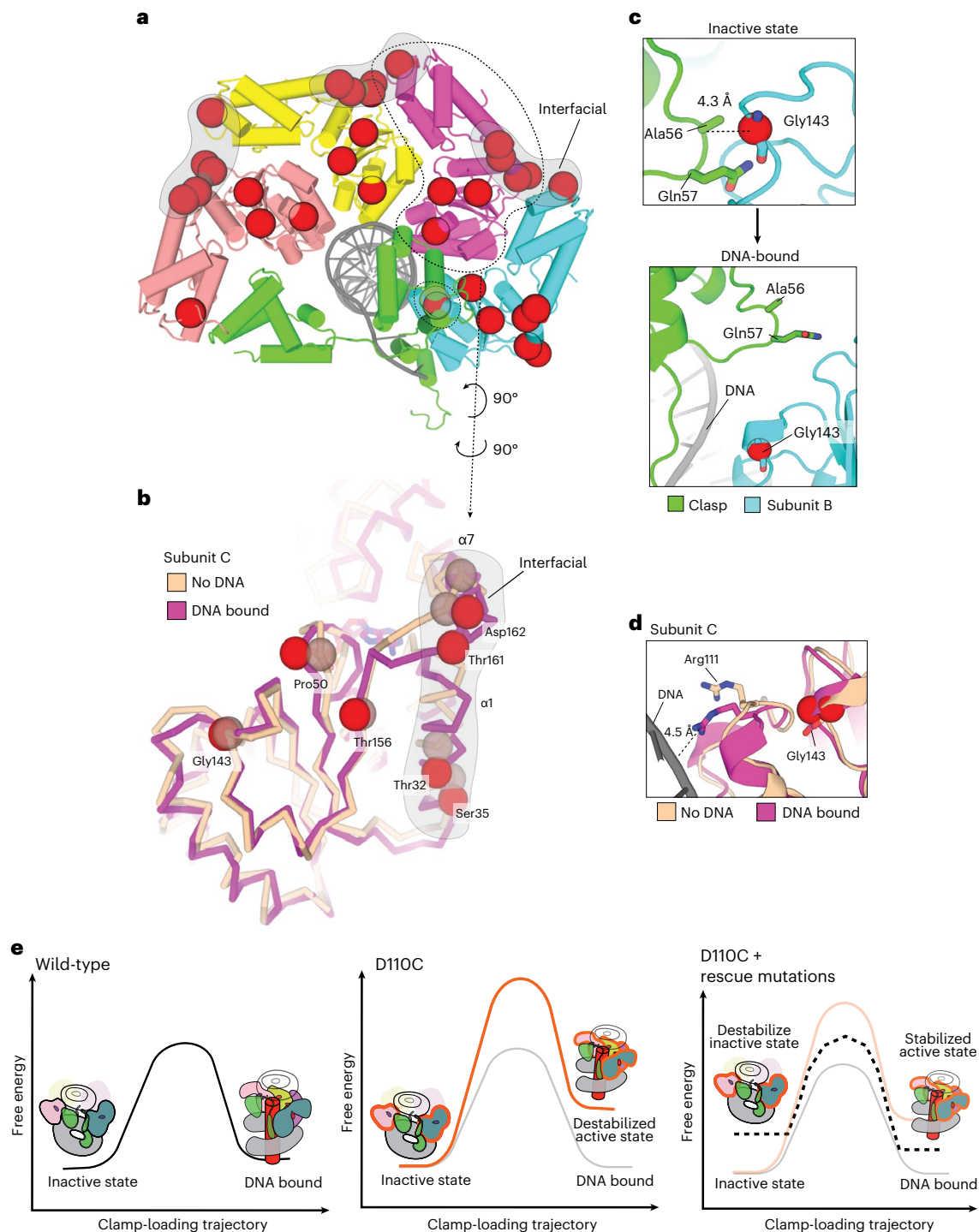


Fig. 6 | p.D110C interrupts the energetic balance between DNA-bound and DNA-unbound states. **a**, The positions of the nine rescue hotspots (red spheres) are mapped onto the active clamp loader cryo-EM structure bound to DNA and clamp. The clasp subunit is colored green, and subunits B–E are colored blue, purple, yellow, and pink, respectively. Two categories of rescue sites are noted. The first category contains sites that sit at the interfaces between subunits, deemed ‘interfacial’ sites (gray areas). The other sites that are not highlighted in gray are sites within the AAA+ module and generally located near the active sites of the ATPase subunits. **b**, Subunit C of the clamp loader complex in **a** (purple) is aligned with the E subunit of the DNA-unbound clamp loader structure (wheat) along the central coupler regions (residues 80–95, 114–127, and 146–152). The alignment is rotated to expose the full interfacial region of the ATPase module (highlighted in gray). The other three sites mapped onto the AAA+ module (Pro50, Gly143, and Thr156) are in regions populated with loops experiencing

internal conformational change between DNA-unbound and DNA-bound structures. **c**, Gly143 (red sphere) within subunit B (cyan) is located near the clasp subunit (green). Top, the inactive, DNA-free structure features close contacts between Gly143 and the clasp subunit residue Ala57. Bottom, aligned on the central coupler region, the DNA-bound cryo-EM structure features displacement of subunit A such that no contacts with Gly143 are made with the clasp subunit. **d**, A focused view of Gly143 is taken from **b**. Gly143 is local with respect to DNA-binding residue Arg111. When DNA is bound, Arg111 is pulled away from Gly143. **e**, The conceptual free-energy landscape of the clamp-loading trajectory is shown for the wild-type, p.D110C mutant, and rescued p.D110C clamp loaders. p.D110C destabilizes the active conformation of the wild-type clamp loader when bound to DNA (orange line). The addition of rescue mutations destabilizes the inactive, DNA-free state of the clamp loader when in the context of p.D110C (dashed line).

structural changes, we aligned the E subunit of the DNA-free state and the D subunit of the DNA-bound state of the cryo-EM structures using the central coupler segment (residues 80–95, 114–127, and 146–152).

There are two key differences between the DNA-bound and DNA-free states of the T4 clamp loader. The first set of conformational differences concerns the overall organization of the active and inactive states, described above. The second set involves changes in the internal structure of the AAA+ modules, depending on whether the module is part of a properly assembled interfacial catalytic center. The internal conformations of the AAA+ modules of subunits B, C, and D of the DNA-bound complex resemble each other closely, and we refer to this as the active conformation of the individual AAA+ modules (Fig. 5b,c). In this active conformation, Asn139 (sensor 1)²⁷, a residue that positions a water molecule for attack on the terminal phosphate of ATP, is located approximately 3 Å away from the BeF₃ moiety. Arg205 (sensor 2)²⁷ interacts with the β phosphate of ADP and with the BeF₃ moiety.

Subunit E of the DNA-bound complex, as well as subunits B and E of the DNA-free complex, is not at a properly configured interface and is in a different conformation, referred to as the inactive conformation of the AAA+ module (Fig. 5b,c). On the basis of the local resolution of the cryo-EM density map of the DNA-free state, subunit E is better resolved than subunit B, likely owing to reduced flexibility within the complex. Because of its higher quality in the cryo-EM density map, we chose to use subunit E (DNA-free, inactive) for our detailed structural comparison with subunit D (DNA-bound, active). The difference between the active and inactive conformations of the AAA+ module is mainly due to a change in the angle between domain 1 (residues 1–120) and domain 2 (residues 160–230) of the AAA+ module. In switching from the active to the inactive conformation, domain 2 rotates by approximately 10° towards domain 1. This leads to a more closed conformation for the AAA+ module in the inactive conformation. The two ATP-sensor residues, Asn139 and Arg205, are pulled away from their positions in the active conformation.

Gly143 is a rescue hotspot in the p.D110C clamp loader, and was identified earlier as a site at which substitutions can compensate for the loss of integrity of the central coupler owing to mutation of Gln118, a structural lynchpin²⁰. The cryo-EM structure of the DNA-free inactive state shows that the Cα atom of Gly143 in subunit B establishes van der Waals contacts with the side chain of Ala56 and the side chain of Gln57 in the A subunit (clasp), with interatomic distances of approximately 4 Å in both cases (Fig. 6c). Alteration of Gly143 is likely to result in steric clashes that increase the energy of the inactive state. When DNA is bound to the clamp loader, the clasp subunit is pulled away from this region, and Gly143 residues in the AAA+ modules point towards DNA but have sufficient space to accommodate substitutions.

Substitution of Gly143 with arginine is particularly activating, consistent with the location of this residue next to DNA in the active complex. Gly143 is located close to Asp110 (the interfacial aspartate residue in the DEAD-box motif) and to Arg111, a residue that is critical for DNA binding. The Cα atom of Gly143 establishes van der Waals contacts with the backbone of Asp110 and Arg111 in both the DNA-bound and DNA-free states. In the DNA-free state, there is a change in the conformation of the backbone, such that Arg111 no longer points towards the DNA (Fig. 6d). This is coupled to a change in the orientation of the Asp110 side chain and a correlated change in Asn139, a catalytically important sensor 1 residue. Given its close proximity to these residues and to DNA, alteration of Gly143 could alter the balance between the active and inactive states.

Several of the hotspot residues are in regions in which structural changes are caused by the rotation of domain 2 of the AAA+ module with respect to domain 1. This rotation is correlated with changes in the backbone within several segments of the protein (the N-terminal segment, residues 1–27, which leads into helix-α1, residues 28–39; the phosphate-binding P loop, residues 48–56; and the loop connecting domain 1 to domain 2, residues 157–163). For example, Pro50 is a rescue

hotspot that is part of the P loop. In the DNA-bound active structure, the carbonyl group of the peptide linkage between Ser49 and Pro50 forms a hydrogen bond with the sensor 1 side chain (Asn139), and this is likely to be important for holding Asn139 in a catalytically competent conformation (Extended Data Fig. 8a). This hydrogen bond is part of a network of interactions made by Asp110 in the DEAD-box motif. In the inactive state, Asn139 is no longer anchored by this backbone interaction. The location of Pro50 at the center of these interactions is consistent with substitutions at this site having an effect on the energetic balance between the inactive and active states.

Helix-α1 contains two rescue hotspots (Thr32 and Ser35). In the inactive DNA-free structure, helix-α1 does not form interfacial interactions (Fig. 6a). However, when domain 2 rotates upon DNA binding, one face of helix-α1 is positioned close to helix-α9 and helix-α10 (residues 212–227) at the B–C and C–D interfaces of the DNA-bound complex, and Thr32 and Ser35 then face the interfacial region. The side chain of Phe28 in helix-α1 packs close to the side chain of Leu227 in helix-α10 in the DNA-bound state (Extended Data Fig. 8b). There is a change in the orientation of helix-α1 in the structure of the DNA-free form, owing to the rotation of domain 2. If this conformation of helix-α1 were maintained when the complex switched to the DNA-bound form, Phe28 would be pulled away from interacting with Leu227. Both residues are highly sensitive to mutation in the p.D110C context, suggesting that the energetics of this interface is important in DNA binding (Fig. 6e).

Discussion

In this work, we studied the mutational sensitivity of a T4 clamp-loader variant with a mild defect in DNA-dependent ATP hydrolysis. This defect was created by replacing the interfacial aspartate residue in the DEAD-box motif of the gp44 ATPase subunit (Asp110) by cysteine (p.D110C). We carried out saturation mutagenesis of the p.D110C variant of the T4 clamp loader and identified rescue hotspots in the AAA+ module, and observed that many different substitutions lead to the recovery of fitness. The structure of the active DNA-bound form of the clamp loader–clamp complex failed to provide explanations for why these mutations lead to increased fitness—most of these are in obscure locations that do not suggest that there is a direct functional relationship.

We hypothesized that there is an as-yet-unidentified inactive state of the T4 clamp loader, and that the rescue mutations alter the energetic balance between the inactive and active states to compensate for the weakening of the interfacial interaction made by Asp110 in the DEAD-box motif. This explanation—involving an inactive state—is more compatible with the observed wide spectrum of mutations at many rescue positions. By contrast, rescue mutations that work by increasing activity directly would be expected to satisfy more rigid structural demands to maintain an active catalytic site, and such mutations would presumably be restricted to only one or two residues at only a few, perhaps conserved, positions.

Cryo-EM analysis of the p.D110C clamp loader confirmed this proposal by revealing a novel DNA-free and autoinhibited state of the T4 clamp loader, in which DNA binding is blocked. This novel autoinhibited state is also seen in cryo-EM reconstructions of the wild-type clamp loader alone and the wild-type clamp loader–clamp complex without DNA or in the presence of ATPγS, an ATP analog. By contrast, the cryo-EM reconstruction of the wild-type T4 clamp loader–clamp–DNA complex in the presence of the ATP mimic ADP–BeF₃ closely resembles the fully active state, as defined previously by X-ray crystallography²². The cryo-EM reconstructions of DNA-bound and DNA-free states of the clamp-loader complex reveal that there are two key conformational changes between the two states. The first concerns the proper assembly of the overall complex. The second concerns internal structural rearrangements within individual AAA+ modules. If there is no neighboring subunit to complete the assembly of a catalytically competent ATPase site, as is the case for subunit E of the DNA-bound state and the subunits

of the DNA-free state, the internal structure of the AAA+ module adopts a more closed inactive conformation. p.D110C rescue mutations occur in regions that undergo conformational changes in transitioning from the inactive DNA-free state to the active DNA-bound state.

Comparison with the mechanism of DEAD-box RNA helicases suggests that clamp loaders have preserved a feature of the RNA helicases, which is to use the DEAD-box motif to couple RNA or DNA binding to ATP hydrolysis. DEAD-box RNA helicases are monomeric proteins that are analogous to dimer units of oligomeric ATPases—each subunit of a monomeric RNA helicase consists of two RecA-type folds, connected by a flexible linker^{28–31}. One of these domains maintains the ability to bind ATP, and the other does not. The ATP-binding domain contains the DEAD-box motif. In the presence of RNA, the second aspartic acid of the DEAD-box motif, corresponding to Asp110 in the T4 clamp loader, coordinates an arginine side chain presented by the other domain, and this interaction stabilizes the formation of a competent catalytic site²⁸. When ATP and RNA are absent, the ion pair between the aspartic acid and the arginine is broken, and the catalytic center is dismantled^{28,32,33}.

The arginine side chain that makes the interfacial contact with the final aspartate in the DEAD box of RNA helicases is structurally analogous to the arginine residue (Arg122) that interacts with the corresponding interfacial aspartate (Asp110) in the T4 clamp loader. Our results suggest that DNA-dependent interfacial interactions made by the DEAD-box motif in the clamp loader provide part of the energy to transition to the fully assembled and catalytically competent state. ATP is integral to the stability of this active state, providing a mechanism for triggering the release of the sliding clamp on DNA once ATP hydrolysis occurs. Release of DNA then causes disassembly of the active sites, which minimizes cycles of futile ATP hydrolysis.

The limited functional constraints on the inactive states of clamp loaders mean that the structures of the inactive states of these machines can diverge dramatically over evolutionary time. This principle is well appreciated in the study of signaling proteins^{34,35}. Over time, this divergence can lead to changes in the sequences that are part of the catalytic center itself, as the nature of the couplings that are required to transition out of the inactive states are altered—the bacterial clamp loaders have lost the DEAD-box motif altogether, although analyses of structures show that they have maintained the same kind of interfacial interactions in this region, but with very different residues. Alterations in the inactive states allow natural selection to tune the responses of these molecular machines, and contribute to the remarkable diversity of ATP-driven machines.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41594-023-01177-3>.

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Methods

Protein expression and purification

The clamp-loader complex (gp44 and gp62 with a C-terminal His $\times$ 6 tag) was prepared as described previously²⁰, with one modification. The cells expressing the p.D110C clamp-loader variant and double mutants containing p.D110C were lysed with a French press (Avestin EmulsiFlex C50 cell homogenizer), using 1,500 p.s.i. pressure at 4 °C for 1.5 to 2 min instead of sonication. Subsequent purification through affinity chromatography was then done as has been described²⁰. The sliding clamp (gp45) was purified as has been described²⁰.

Preparation of primer-template DNA

The DNA oligonucleotides were obtained from Integrated DNA Technologies (IDT) and had the following sequences:

Template strand (30 bp): 5'-TTTTTTTTTTTATGTACTCGTAGTGTCTG-3'

Primer strand (20 bp): 5'-GCAGACACTACGAGTACATA-3'

DNA oligonucleotides were ordered as lyophilized samples and diluted with 20 mM Tris (pH 7.5) and 1 mM MgCl₂ to a final concentration of 200 μ M. For fluorescence anisotropy measurements, the template strand had a TAMRA fluorophore attached at the 5' end. Primer and template DNA were mixed in an equimolar ratio, brought to 95 °C, and cooled to room temperature over a period of 2 h.

Fluorescence anisotropy DNA-binding assay

Primer template DNA (50 nM) labeled at the 5' end of the template with a TAMRA fluorophore (purchased from IDT) was incubated in buffer A (100 mM HEPES pH 7.5, 150 mM NaCl, 10% (vol/vol) glycerol), with 1 mM ADP-BeF_x (1 mM ADP dissolved in 20 mM HEPES pH 7.5, 1 mM BeSO₄, 5 mM NaF), 5 mM MgCl₂, and 7 μ M sliding clamp (gp45). When preparing ADP-AlF_x, 1 mM BeSO₄ was replaced with 1 mM AlCl₃. ATP γ S was purchased from Jena Biosciences. Clamp loader was serially diluted in buffer A and titrated in the above reaction mixture, ranging from 0–5 μ M for a total reaction volume of 100 μ L. After a 3-h incubation at room temperature (Extended Data Fig. 9)^{38,39}, fluorescence polarization was measured by a Biotek Synergy H1 plate reader. Apparent K_d values were estimated by nonlinear curve fitting to the Hill equation using GraphPad Prism 9.

ATP hydrolysis assay

The ATP hydrolysis rates of the various gp44 and gp62 clamp-loader complexes were measured using a coupled enzyme assay in which the decrease in absorbance at 340 nm was monitored as NADH was converted to NAD⁺ during the regeneration of the ADP product of the ATP hydrolysis reaction⁴⁰. Each well's regeneration system contained 3 mM phosphoenolpyruvate, 0.5 mM NADH, and 1,079 units per ml of pyruvate kinase and lactate dehydrogenase (Sigma Aldrich). The reaction buffer contained 25 mM HEPES (pH 7.5) and 9 mM magnesium acetate⁴¹. The reaction also included 0.2 μ M gp44/gp62 complex, 1.5 μ M gp45, and 1.5 μ M primed DNA per well. To initiate the reaction, ATP was added as a 1:1 ATP:magnesium acetate solution, for a final concentration of 100 μ M ATP in each 50- μ L sample, and the assay temperature was maintained at 23 °C. Absorbances were read every 30 s for 40 min in a Biotek Synergy H1 plate reader. The molar absorption coefficient for NADH used was $K_{\text{path}} = 6,220 \text{ M}^{-1}\text{cm}^{-1}$.

Crystal structure of the p.D110C T4 clamp loader mutant bound to DNA and the sliding clamp

For crystallization, the p.D110C mutant clamp loader (6 mg ml⁻¹), clamp, and DNA were mixed in a buffered solution (20 mM Tris pH 7.5 and 100 mM NaCl) to a molar ratio of 1:2.7:2.7 immediately before crystal trays were set. ADP-AlF_x was formed within this solution by the addition of ADP (4 mM), NaF (13 mM), AlCl₃ (4 mM), and MgCl₂ (20 mM). Crystals were obtained through hanging drop diffusion (1 μ L protein complex solution mixed with 1 μ L reservoir solution) with 0.1 M MES pH

6.5, 9% PEG MME 5,000, and 6% 1-propanol reservoir solution (1 ml). We attempted to crystallize the p.D110C mutant clamp loader–DNA–clamp complex with ADP-BeF_x, ADP-AlF_x, and ATP γ S. Only the use of ADP-AlF_x produced crystals.

For data collection, crystals were soaked with well solution with 10–20% (vol/vol) glycerol, which served as a cryoprotectant. Cryogenic diffraction data were collected at the Advanced Light Source (Lawrence Berkeley National Lab, Beamline 5.0.1. with an X-ray wavelength of 0.9998 Å). Diffraction data were processed using XDS⁴² and AIMLESS⁴³ and star-aniso⁴⁴, to a high resolution cutoff of 3.10 Å. Molecular replacement was performed with Phaser⁴⁵, as implemented in the Phenix suite⁴⁶ using a combination of PDB 3U60 and 3U5Z as a search model. Refinement was performed using Phenix and model building was done in Coot⁴⁷. Non-crystallographic symmetry restraints between ATPase subunits were not imposed (see below). The p.D110C mutant clamp-loader structure and data are deposited at PDB 8UK9.

X-ray crystallography: re-refinement of wild-type clamp loader structure with DNA and clamp

Because we observed structural changes to the ATPase subunits when they were not bound to DNA, we sought to locate any internal structural changes of the ATPase domain when DNA was bound to the clamp loader. Unfortunately, the imposed non-crystallographic symmetry restraints masked any unique conformational changes between subunits in our previously published T4 clamp-loader structure. Thus, we reprocessed data from the original T4 bacteriophage clamp loader–clamp–DNA structure deposition (PDB 3U60), without non-crystallographic symmetry restraints, with XDS⁴² and AIMLESS⁴³. The original diffraction data also contained high anisotropy, so we also used the star-aniso web server⁴⁴ to improve the diffraction data processing. This resulted in data with a higher-resolution cut-off (2.63 Å). From the re-refinement, the free R value improved from 0.273 to 0.244, and the stereochemical parameters for the model improved. The newly processed data and refined structure are deposited at PDB 8UH7.

Phage-propagation assay

The phage-propagation assay was carried out as described previously²⁰. Briefly, T4 bacteriophage engineered to lack clamp-loader genes (T4^{del}) was used to infect *E. coli* cells (BL21) harboring the CRISPR–Cas12a system and a plasmid from the constructed libraries. Before infection, a sample of the cells containing the libraries was saved for DNA sequencing as the 'input' clamp loader gene population. After a phage replication period of approximately 14–16 h, the phage present in the culture supernatant were subject to PCR amplification of the recombinant clamp-loader locus. The clamp-loader genes present in the input cell sample were also amplified using PCR. Both sample types were then sequenced using Illumina 500 cycle kits with an Illumina Miseq.

Construction of the saturation mutation library of Asp110 within the ATPase subunit (gp44)

The wild-type clamp-loader helper plasmid used in previous work was used as the base for the introduction of a single degenerate site (codon sequence NNS) at position 110 of T4 bacteriophage gp44 using Golden Gate cloning methods, as previously described²⁰. Owing to the small library size (23 codons introduced by NNS degenerate primers), library coverage at every stage of library generation was well over 1,000-fold. This plasmid was used to transform *E. coli* (BL21 cells) and was subjected to the phage-propagation assay.

Saturation mutagenesis of the D110C T4 clamp-loader variant

To create the library for saturation mutagenesis of the AAA+ module of the ATPase subunit (gp44), the wild-type saturation mutagenesis library generated previously²⁰ was used, in which every residue in the AAA+ module was replaced by each of the 20 amino acids, plus the stop codon, one at a time.

The library was created in two pools. In one pool (pool A), residues 2–118 were varied, one mutant per plasmid. Residues 116–230 were varied in the second pool (pool B). The D110C mutation was introduced on top of this existing set of mutations using primer-guided mutagenesis. Primers with an annealing area spanning from 23 nucleotides downstream of position 110 to 32 nucleotides upstream of position 110 sensing the wild-type sequence were used to introduce a mutagenetic codon for cysteine at position 110. The 5' ends of these primers contained a recognition motif for the BsaI restriction enzyme for Golden Gate cloning⁴⁸. As result, residues 2–118 of pool A were varied, except for residues 101–115, which retained the wild-type identity as they were sites of primer annealing for the introduction of the D110C mutation. Every residue of pool B was varied. Library diversity during the cloning stages was maintained at >20× and was assessed by colony counts of transformants. These two library pools were used to transform *E. coli* (BL21) and were subjected to the phage-propagation assay.

Cryo-EM sample preparation and data acquisition

To assemble the ternary complex of clamp loader–clamp–DNA, clamp loader, clamp, and primed DNA were mixed at a molar ratio of 1:1.2:1.2 to a final protein concentration of 5–10 mg ml⁻¹. ADP, NaF, BeCl₂, MgCl₂, and *n*-octyl-β-D-glucopyranoside (β-OG) were added to a final concentration of 1 mM, 10 mM, 2 mM, 10 mM, and 0.015% (wt/vol), respectively. To assemble the binary complex of clamp loader–clamp, clamp loader and clamp were mixed at a molar ratio of 1:1.2, to a final protein concentration of 5–10 mg ml⁻¹. ATPγS, MgCl₂, and β-OG were added to a final concentration of 1 mM, 10 mM, and 0.015% (wt/vol), respectively. The initial cryo-EM trials for these complexes were unsuccessful, owing to severely restricted orientations of the particles on the cryo-EM grids. Addition of the detergent β-OG at a concentration of 0.015% (wt/vol) during grid preparation eliminated the preferred orientation, enabling single-particle reconstruction of the T4 clamp loader–clamp–DNA complex at a resolution of approximately 3.2 Å.

For cryo-EM grid preparation, QUANTIFOIL R 1.2/1.3 300-mesh gold grids were made hydrophilic by glow-discharge using the PELCO easiGlow system for 45 s at 25 mA. After 3.5 μl of clamp-loader complexes was applied, grids were plunge-frozen using a Vitrobot Mark IV (Thermo Fisher Scientific) with a blotting time of 2.5 s, blotting force of 20, chamber temperature of +10 °C, and humidity of 95%. The grids were imaged using a Titan Krios G2 transmission electron microscope (Thermo Fisher Scientific), operated at 300 kV. Images were recorded on a K3 camera (Gatan), operated in super-resolution counting mode with a physical pixel size of 0.524 Å. The detector was placed at the end of a GIF Quantum energy filter (Gatan), operated in zero-energy-loss mode with a slit width of 20 eV. All videos were collected at a nominal magnification of ×81,000 and a defocus range between −0.8 and −2.2 μm., using the automated data collection software SerialEM⁴⁹. The nominal dose per frame was approximately 1 e⁻ Å⁻², and the total dose per video is approximately 50 e⁻ Å⁻². For the clamp loader–clamp–DNA complex, a total of 10,508 videos were collected. For the clamp loader–clamp complex, a total of 18,727 videos were collected.

EM image processing

Cryo-EM images were processed using cryoSPARC v4 (ref. 50). For the clamp loader–clamp–DNA complex, 10,508 videos were motion-corrected using patch-based motion correction. After local CTF determination using patch-based CTF estimation, a total of 4,206,415 particles from 10,508 micrographs were automatically selected using template-free Blob_picker and were extracted from the micrographs with a binning factor of 2. A relatively low threshold value was used during the particle picking to include particles with less contrast. More than half of the initial picks are from the background. After iterative two-dimensional (2D) classification, most of the background particles were removed on the basis of 2D class averages. The remaining 2,019,311 particles were further processed by ab initio reconstruction into four classes. The resulting

3D maps showed that three of the ab initio reconstruction classes were mostly 'junk' particles without recognizable structural features, and the other class showed clear resemblance to the previously reported crystal structure of the clamp loader–clamp–DNA complex (PDB 3U60). This class of particles (754,126) was selected and re-extracted from the micrographs without binning. Another round of ab initio reconstruction was performed to sort the remaining particles into two classes: a 'junk' class (200,803 particles) without any recognizable structural feature, and a 'clean' class (553,323 particles) resembling the clamp-loader complex. The remaining particles were subjected to iterative rounds of heterogeneous refinement with two classes, with each round resulting in a class of particles that yielded improved map quality relative to the previous round. After the map quality stabilized, the remaining 204,448 particles were subjected to a final round of non-uniform refinement, with the Defocus Refinement option turned on. The resulting map (overall resolution, 3.15 Å) was used for model building and refinement.

For the clamp loader–clamp complex, 18,727 videos were motion-corrected using patch-based motion correction. After local CTF determination using patch-based CTF estimation, a total of 17,937,844 particles from 18,727 micrographs were automatically selected using template-free Blob_picker and were extracted from the micrographs with a binning factor of 2. A low threshold value was used during the particle picking to include particles with less contrast. After iterative 2D classification, 2D classes with no recognizable feature were removed. Three-dimensional ab initio reconstruction was performed on two subsets of particles to generate the initial models of the sliding clamp and clamp loader–clamp complex. The first subset of particles (711,674) contains 2D classes showing only the sliding clamp. The second subset of particles (314,348) contains 2D classes showing the clamp loader–clamp complex. The resulting models of the sliding clamp and clamp loader–clamp complex, together with a 'junk' model generated from background particles, were used as input models for one round of heterogeneous refinement to sort the remaining 9,692,266 particles into three classes. Particles corresponding to the clamp loader–clamp class (2,525,436) were selected and re-extracted from the micrographs without binning. One round of heterogeneous refinement was performed to sort the remaining particles (2,525,436) into ten classes. After visual inspection, the class of particles (455,041) corresponding to the map with best quality was selected and subjected to a final round of non-uniform refinement, with the defocus refinement option turned on. The resulting map (overall resolution: 3.21 Å) was used for model building and refinement.

Model building and refinement

To generate the initial model for the cryo-EM structure of the clamp loader–clamp–DNA complex (DNA-bound state), the re-refined crystal structure of the T4 clamp loader–clamp–DNA complex (PDB 8UH7) was rigid-body fitted into the reconstructed cryo-EM map of the clamp loader–clamp–DNA complex, using UCSF Chimera⁵¹. The UCSF Chimera-fitted model was subjected to one round of real-space refinement⁵² in Phenix. The model was then manually adjusted in Coot, followed by iterative rounds of real-space refinement in Phenix and manual fitting in Coot. Model quality was assessed using the comprehensive model validation tools from Phenix.

To generate the initial model for the cryo-EM structure of the clamp loader–clamp complex (DNA-free state), the re-refined crystal structure of the T4 clamp loader–clamp–DNA complex (PDB 3U60) was split into five fragments: the pentameric collar domains (residues 233–319 of subunits B, C, D, and E, and residues 40–106 of subunit A), the A' leg (residues 2–39, subunit A), the A' leg (residues 107–187, subunit A), the AAA+ module of subunit B (residues 1–232, subunit B), and the AAA+ module of subunit E (residues 1–232, subunit E). The structure of a closed sliding clamp was taken from the previously reported crystal structure of the T4 clamp loader–DNA complex bound to a closed sliding clamp (PDB 3U61, chain F, G, and H). These six fragments (five from PDB 3U60, and one from PDB 3U61) were rigid-body fitted into

the reconstructed cryo-EM map of the clamp loader–clamp complex, using UCSF Chimera. The Chimera-fitted models were then each subjected to one round of real-space refinement in Phenix. The six models of fragments were manually adjusted and merged into one model in Coot, followed by iterative rounds of real-space refinement in Phenix and manual fitting in Coot. Model quality was assessed using the comprehensive model validation tools from Phenix. All maps and models were visualized using UCSF Chimera and PyMOL.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Deep mutagenesis data generated during this study have been deposited in the Sequence Read repository with BioProject accession number [PRJNA1025274](#). Deep mutagenesis data of the wild-type clamp loader can be found in our previous publication²⁰. Crystal structures of the p.D110C mutant bacteriophage clamp loader–clamp–DNA complex and reprocessed wild-type bacteriophage clamp loader–clamp–DNA complex have been deposited in the PDB with accession codes [8UK9](#) and [8UH7](#). The cryo-EM density maps of T4 bacteriophage clamp loader–clamp DNA-bound complex and T4 bacteriophage clamp loader–clamp DNA-free complex been deposited to the Electron Microscopy Data Bank under the accession codes [EMD-42399](#) and [EMD-42402](#). The associated coordinates have been deposited to the PDB under accession codes [8UNF](#) and [8UNH](#). Biochemical data are supplied as Excel spreadsheets, with relevant figures listed in the filename. Source data are provided with this paper.

Code availability

Bash scripts used to extract counts per allele from the raw sequencing data have been deposited in Github: <https://github.com/KuriyanLab/DMS-D110C-NSMB2023>

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Acknowledgements

We thank T. Eisen for assistance in the review of this manuscript and helpful discussions about improving our data analysis. We thank the Berkeley Center for Structural Biology beamline staff at the Advanced Light Source, Lawrence Berkeley National Laboratory. We also acknowledge the Berkeley Bay Area Cryo-EM Facility for cryo-EM data collection. Molecular graphics and analyses performed with UCSF Chimera, developed by the Resource for Biocomputing, Visualization, and Informatics at the University of California, San Francisco, with support from NIH P41-GM103311. Funding for this work was provided by the Howard Hughes Medical Institute and NIH R01-GM144512.

Author contributions

K.M., Y.H., and J.K. conceived the project; K.M. and S. Subramanian designed the high-throughput experiments; K.M. performed and analyzed the high-throughput experiments; Y.H. and S. Subramanian performed and analyzed the EM experiments; K.M., K.G., S.G.-K, X.L.R., and L.Z. designed and performed the biochemical assays; K.M., C.G., and S.G.-K. performed and analyzed the X-ray crystallography experiments; J.K. supervised the project; K.M., Y.H., and J.K. drafted the manuscript; K.M., Y.H., S. Subramanian, and M.O'D. edited the manuscript. All authors commented on the manuscript.

Competing interests

J.K. is a cofounder of Nurix Therapeutics and is on the scientific advisory boards of Carmot and Revolution Medicine. S. Subramanian is founder and chief executive officer of Gandeveva Therapeutics. The remaining authors declare no competing interests.

Additional information

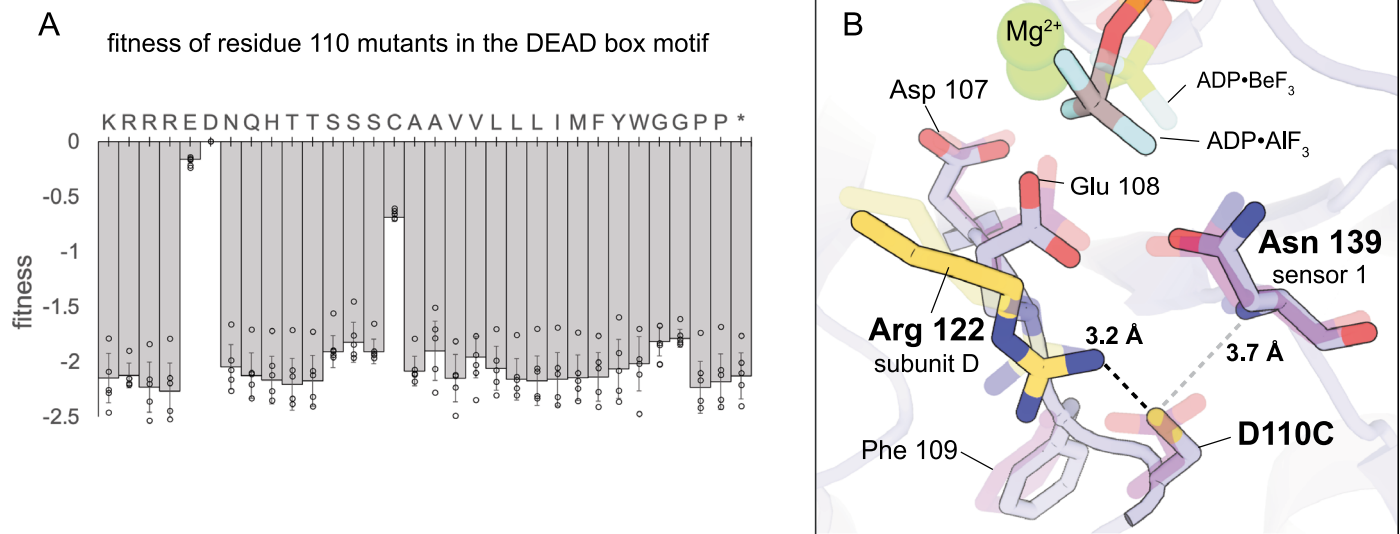
Extended data is available for this paper at <https://doi.org/10.1038/s41594-023-01177-3>.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41594-023-01177-3>.

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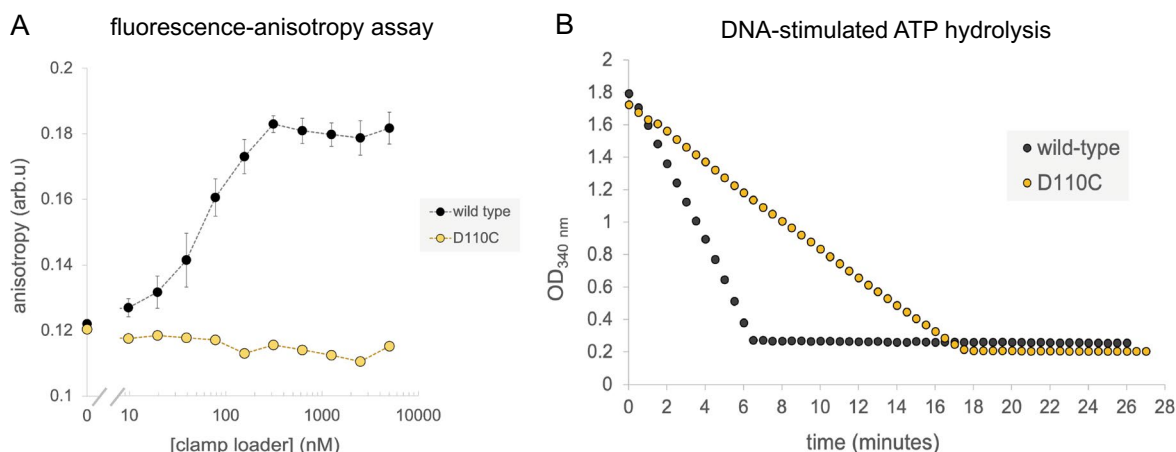
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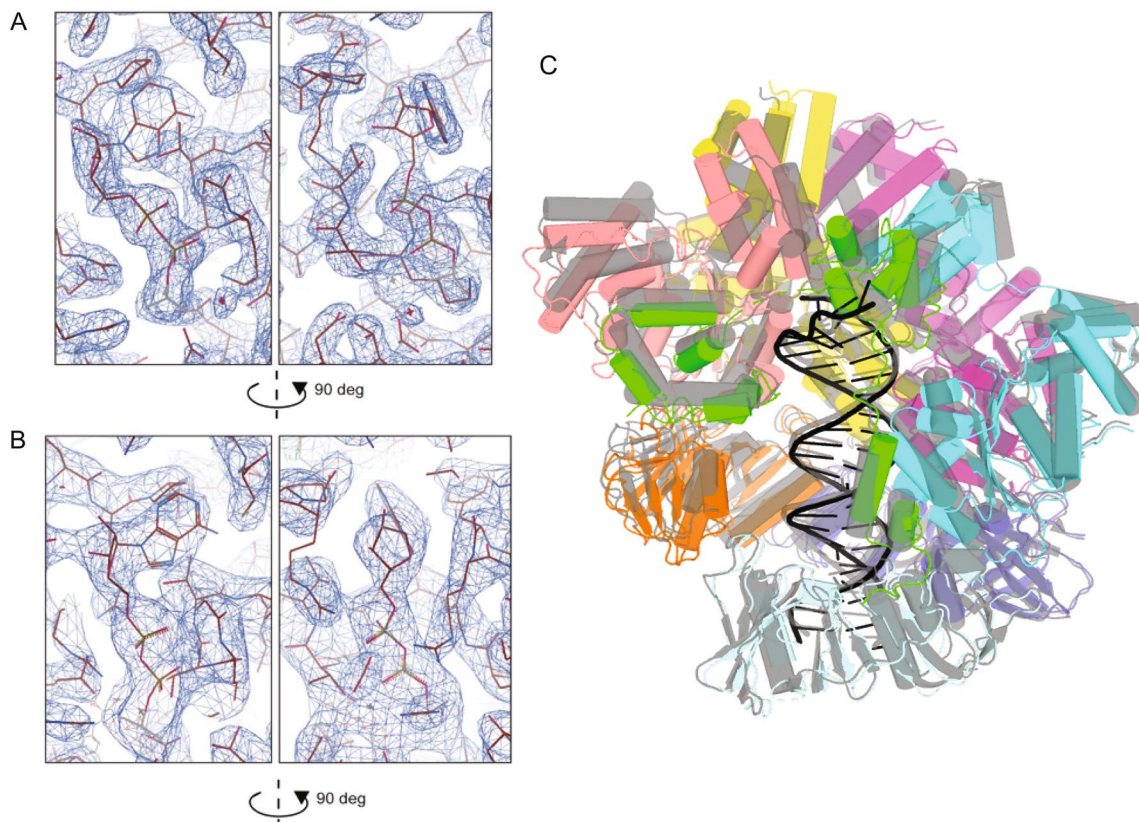
Extended Data Fig. 1 | Mutational tolerance and location of position 110 within T4 bacteriophage clamp loader AAA+ ATPase domains. **a**, We used a focused mutagenesis library in which only the identity of the residue at position 110 was varied and measured the fitness of the variants. The fitness assay was conducted as described in Fig. 1b. Fitness of each substitution of position 110 is plotted. The fitness of synonymous codons was also measured to insure internal reproducibility of fitness measurements. Data are presented as mean values $\pm$ standard deviations (SD), $n = 5$ biologically independent experiments **b**, The crystal structures of the remodeled wild-type clamp loader (translucent) and D110C mutant clamp loaders are aligned on residues 1–120 (domain I) of subunit C (purple). At each of the ATP-binding sites in the wild-type complex, the sidechain of Asp 110, in the DEAD-box motif, forms hydrogen bonds with

an arginine sidechain (Arg 122) provided by the adjacent subunit, and also with the backbone of the catalytically important sensor 1 residue, Asn 139²⁷. In the D110C complex, the cysteine residue at position 110 interacts with only one of these residues at a time. When the cysteine sidechain does interact closely with that of Arg 122, the termini of the sidechains are within 3.0 Å of each other. This suggests that the cysteine is deprotonated and negatively charged at those sites, explaining the mild phenotype of the D110C mutant. The inability of the cysteine sidechain to engage Arg 122 and Asn 139 simultaneously results in a subtle change in the interfacial geometry between subunits that is distributed throughout the structure. This leads to a slight change in the overall engagement with DNA, which might result in the weakened DNA affinity.



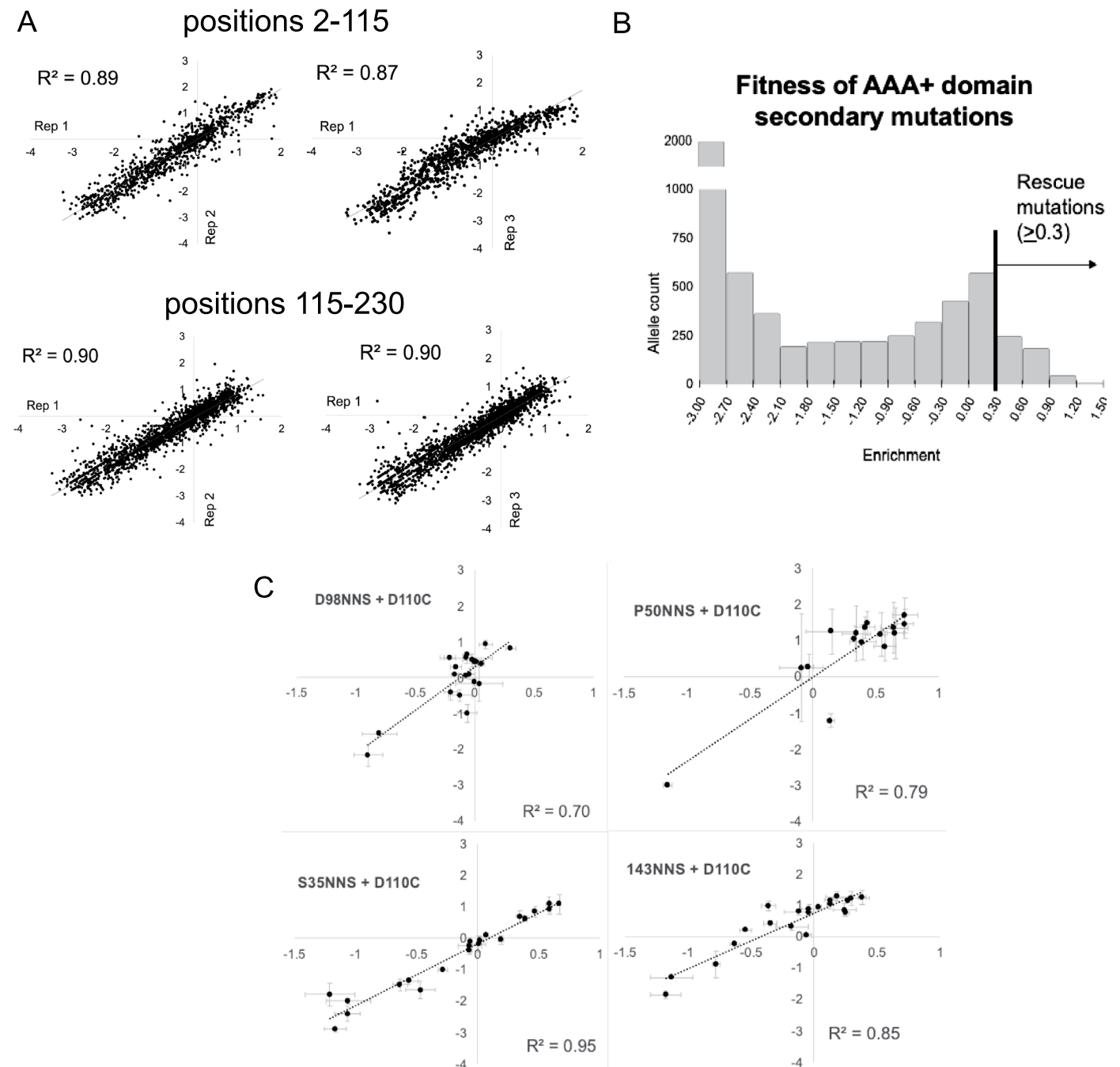
Extended Data Fig. 2 | Biochemical assessment of D110C mutant clamp loader. **a**, Purified wild-type T4 clamp-loader complex and the D110C variant are assessed for DNA-binding affinity by using a fluorescence polarization anisotropy assay. In previous work, a fluorescent tag on a DNA duplex with a 5' overhang (representing primed DNA) was used to measure the change in fluorescence anisotropy of the labeled DNA after adding the clamp loader^{22,53,54}. In the present experiments, we used primed DNA (50 nM) consisting of a 20 base-pair DNA duplex with a 10-nucleotide overhang labeled with TAMRA dye. The labeled DNA was added to a reaction mixture containing the T4 sliding clamp (7 μ M) and the ATP analog ADP-BeF₃ (1 mM). Purified clamp loader (0–5 μ M) was then titrated into the sliding-clamp:primed DNA:nucleotide analog mixture and allowed to equilibrate for 3 hours at room temperature before fluorescence anisotropy measurement (see Methods). The anisotropy per clamp

loader concentration is plotted for wild-type D110C mutant clamp loader. Data are presented as mean values $\pm$ standard deviations (SD), $n = 3$ biological replicates. **b**, We measured the kinetics of ATP hydrolysis by using an assay in which the consumption of ATP by the clamp loader is coupled to the oxidation of NADH^{12,40,54}. Kinetic measurements were made for both the wild-type clamp loader and the D110C mutant (0.5 μ M) bound to clamp (2.5 μ M) and the same primed DNA construct used for the fluorescence anisotropy assay (2.5 μ M), but without the fluorophore label. ATP hydrolysis rate of the D110C mutant is ~33% of the rate exhibited by the wild-type clamp loader. The DNA concentration used for this assay is ~5-fold higher than the K_D value for the wild-type clamp loader, which presumably accounts for the partial stimulation of ATP hydrolysis by the D110C mutant. Example traces of these kinetic measurements are shown.



Extended Data Fig. 3 | Electron density of wild-type and D110C T4 bacteriophage crystal structures. **a**, The electron density map (2FoFc) of the re-refined wild-type clamp loader structure is shown centered around the active site of subunit D. The map is rendered at a contour level of 0.6176 e/Å³ (2.09 RMSD). Two view angles of the active site are shown. The nucleotide is the central axis of the view. **b**, The electron density map (2FoFc) of the D110C clamp loader mutant structure is shown centered around the active site of subunit D. The map

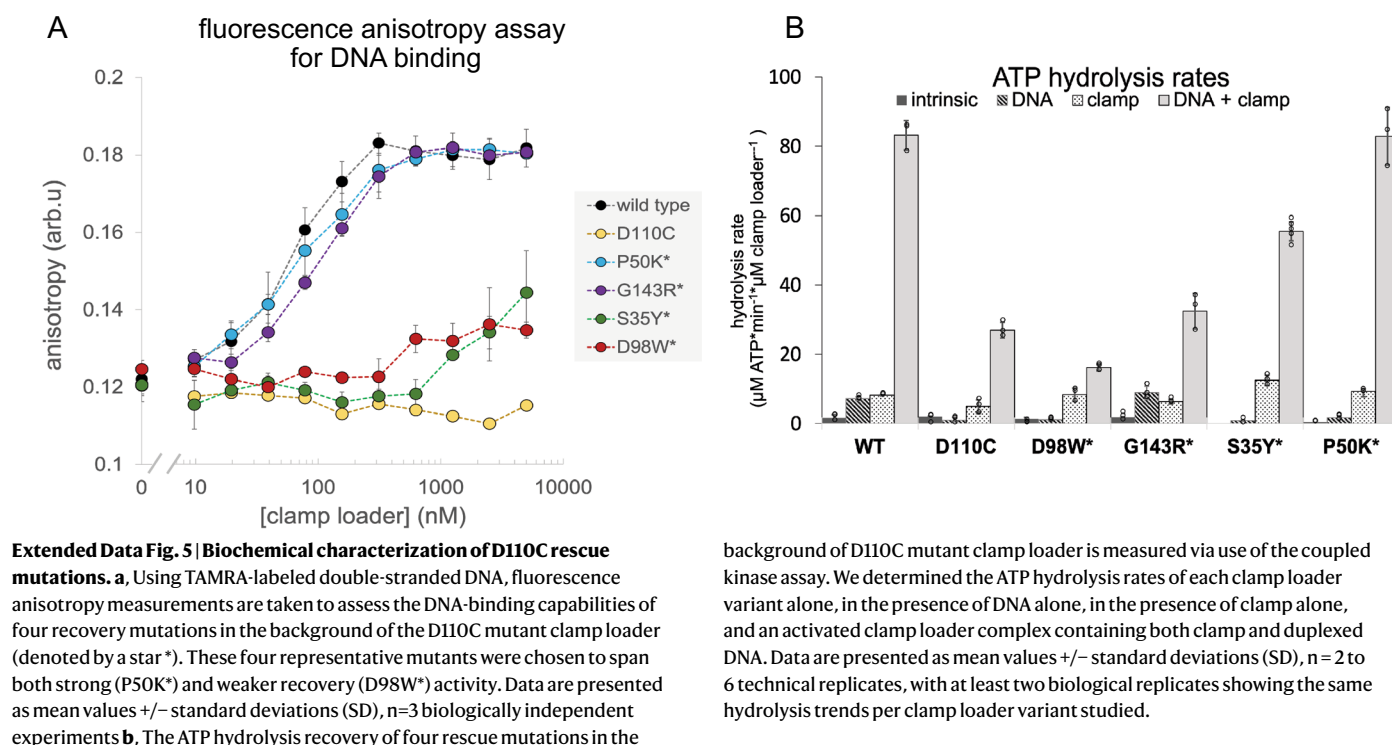
is rendered at a contour level of 0.2253 e/Å³ (2.07 RMSD). Two view angles of the active site are shown. The nucleotide is the central axis of the view. **c**, The crystal structure of D110C mutated clamp loader (molecule A) is aligned against the wild-type clamp loader structure (gray). The alignment is performed on subunits C and D of both structures. The A, B, C, and E, subunits of the D110C structure are colored green, blue, magenta, yellow, and pink, respectively. The clamp subunits of the D110C structure are colored light blue, purple, and orange.

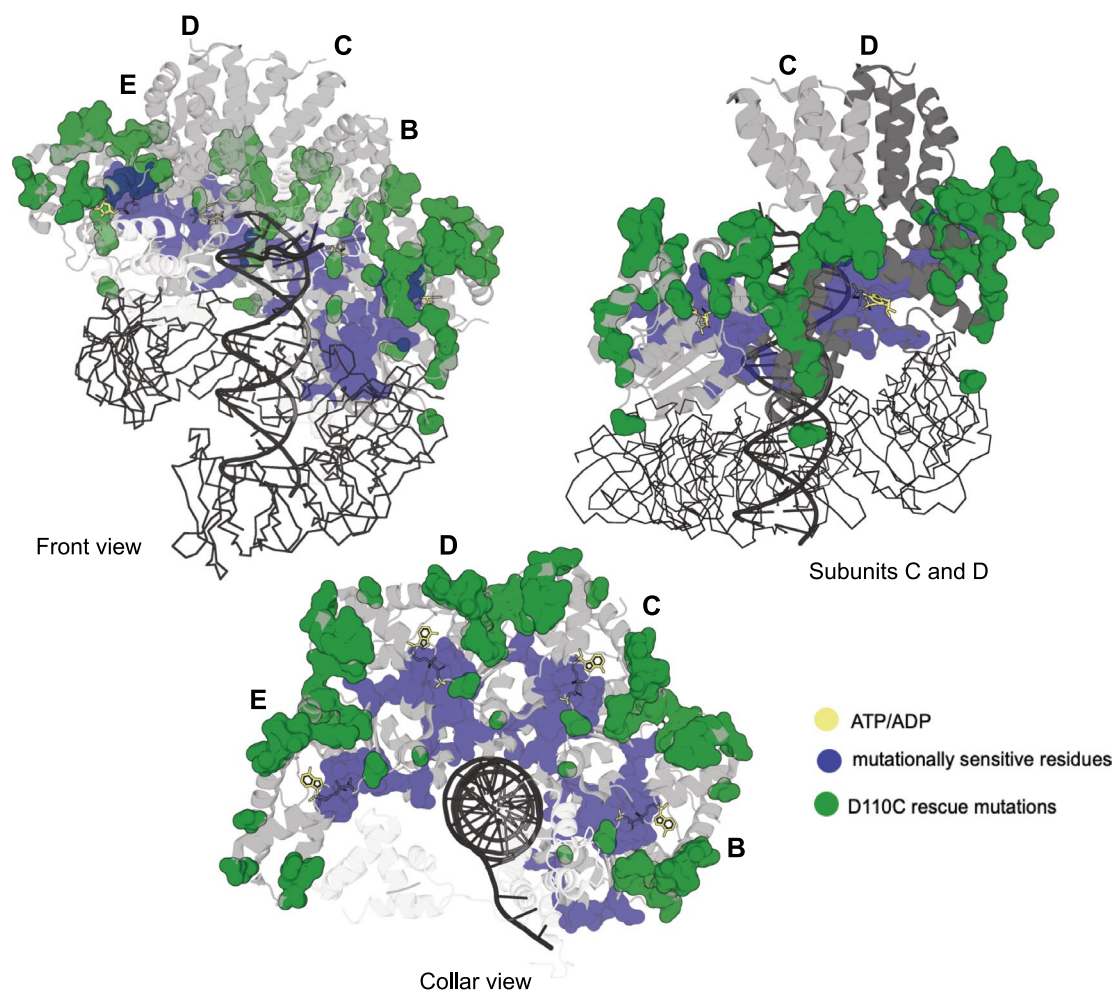


Extended Data Fig. 4 | Statistics of D110C clamp loader deep mutagenesis screen.

a, Agreement between fitness measurements from three replicate experiments per assay pool (residues 2-115 assay pool $n=1130$, and residues 115-230 assay pool $n=2076$) is shown in scatter plots. Each point in the scatter plot represents the fitness measurements made from the two trials. **b**, The histogram shows the spread of relative fitness values for the AAA+ module of the ATPase

subunit in the background of the D110C mutant clamp loader. **c**, Small-scale libraries of recovery mutations in the background of D110C were constructed and subjected to the phage replication assay. The fitness of this small library per substitution (x-axis) is plotted against the fitness scores of same mutations as assayed in the full-scale deep mutagenesis screen (y-axis). Data are presented as mean values $\pm$ standard deviations (SD), $n=3$ biological replicates.

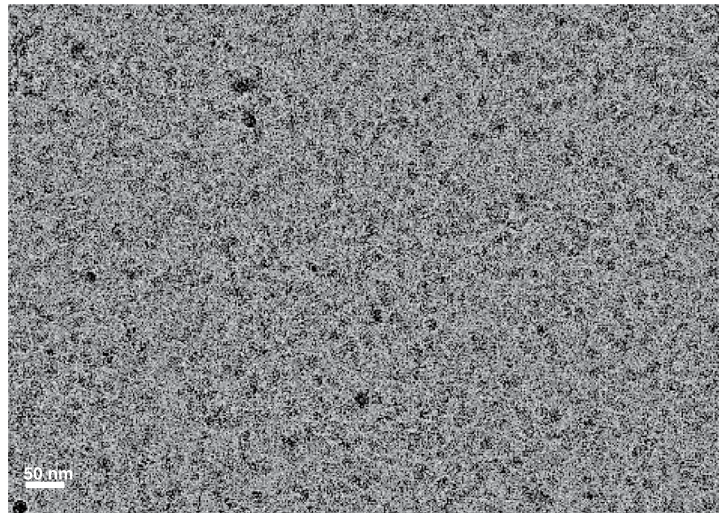




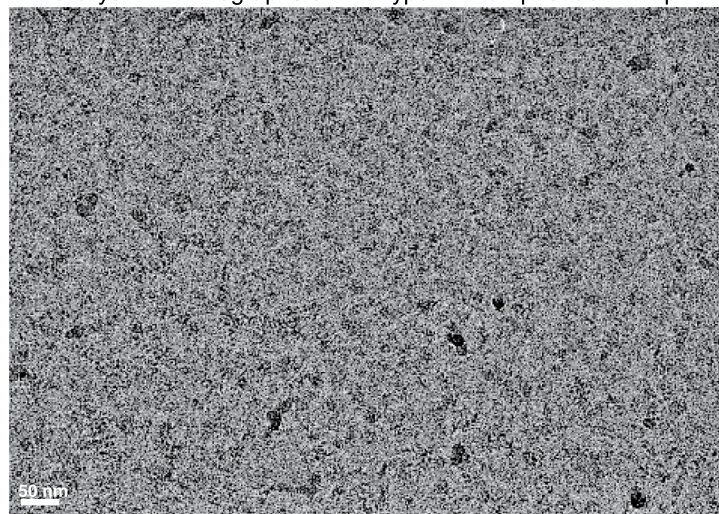
Extended Data Fig. 6 | Mapping recovery mutations onto structure with wild-type sensitive residues indicated. Recovery positions having at least 6 substitutions that recover D110C replication fitness are mapped onto the

wild-type clamp loader crystal structure (PDB ID: 3U60) in green. Positions that are identified to be particularly sensitive in the wild-type background (Fig. 2) are shown in blue.

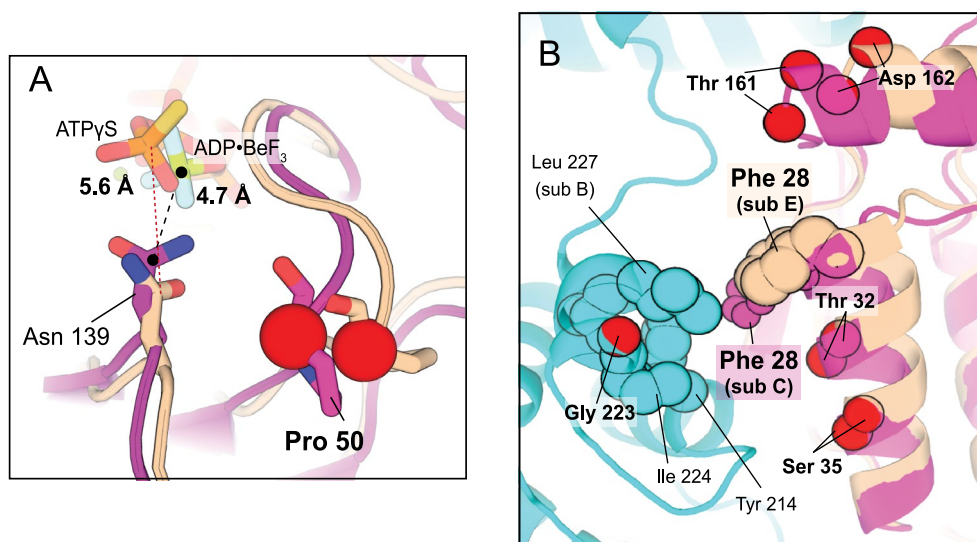
A Representative Cryo-EM micrographs of wild-type T4 clamp-loader:clamp with DNA



B Representative Cryo-EM micrographs of wild-type T4 clamp-loader:clamp without DNA



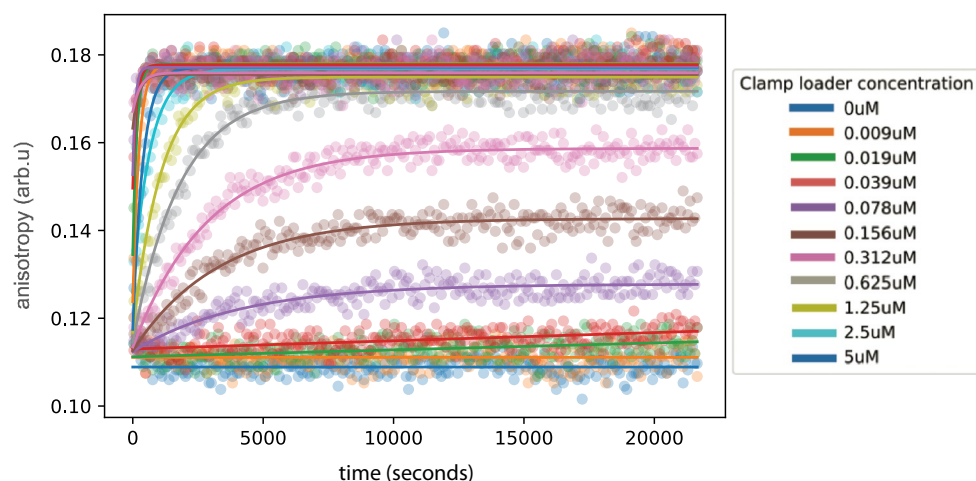
Extended Data Fig. 7 | Representative Cryo-EM micrographs. **a.** Representative Cryo-EM micrographs of wild-type T4 clamp-loader:clamp with DNA **b.** Representative Cryo-EM micrographs of wild-type T4 clamp-loader:clamp without DNA.



Extended Data Fig. 8 | Mechanistic evaluation of rescue hotspot residues.

a, An expanded view from Figure 7b is taken of residue Pro 50 (red spheres). The DNA-unbound structure (wheat) features greater distance between Pro 50 and Asn 139. The distance between ATPyS and Asn 139 in this structure is also greater (red dashes, 5.6 angstroms). When DNA is bound (purple), Pro 50 moves closer to Asn 139, while Asn 139 moves closer to ADP·BeF₃ (black dashes, 4.7 angstroms)

b, The interfacial surface seen in Fig. 7a is expanded to show five distinct rescue hotspots (red spheres) around the hydrophobic plug residue, Phe 28. The DNA-bound conformation of chains C (purple) and B (blue) is aligned with the DNA-unbound conformation of chain C (wheat) along the central coupler region. The hydrophobic plug is formed by residues Tyr 214, Ile 224, and Leu 227.



Extended Data Fig. 9 | Kinetic measurement of fluorescence anisotropy over a range of clamp loader concentrations. The fluorescence anisotropy of clamp loader to fluorescently labeled primer-template DNA is measured over time and across clamp loader concentration. This complex was also bound to the

non-hydrolysable analog ADP*BeF₃. Very slow equilibration of the anisotropy readout is seen at low clamp loader concentration (between 0.009 uM and 0.039 uM). Clamp loader at a concentration of around 0.08 uM generates consistent anisotropy values after approximately 10,000 seconds (~2.5 hours).

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Data collection	BioTek Gen5 for collection of spectroscopic data from a BioTek H1 plate reader, Automated data collection on the Titan Krios was performed using serialEM 3.7
Data analysis	Matlab R2022b, MotionCor2, Cryosparc v4, Phenix 1.19.2, Coot 0.8.9.2, PyMOL 2.2.3, UCSF Chimera 1.13

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Deep mutagenesis data generated during this study have been deposited in the Sequence Read repository with BioProject accession number PRJNA1025274. Mutagenesis data processing information and protocols are provided as supplemental files. Crystal structures of the D110C mutant bacteriophage clamp-loader:clamp:DNA complex and reprocessed wild-type bacteriophage clamp-loader:clamp:DNA complex have been deposited in the PDB with accession codes 8UK9

and 8UH7. The cryo-EM density maps of T4 bacteriophage clamp-loader:clamp DNA-bound complex and T4 bacteriophage clamp-loader:clamp DNA-free complex, have been deposited to the Electron Microscopy Data Bank (EMDB) under the accession codes EMD-42399 and EMD-42402. The associated coordinates have been deposited to the Protein Data Bank (PDB) under accession codes 8UNF and 8UNH.

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Sample size

Libraries constructed for deep mutagenesis screens ranged from 32 to 3600 mutants. Library coverage within transformed *E. coli* prior to selection was assessed via next-generation sequencing and was deemed to be sufficient for each of the reported experiments. No statistical tests were used to predetermine sample sizes. For the cryo-EM samples, sample sizes were chosen based on established practice in the field and were determined by available electron microscopy time and density of particles on electron microscopy grids. The sample size is sufficient to obtain structures at the reported resolution. For the clamp-loader:clamp:DNA complex, 10,508 movies were collected and 204,448 particles were sufficient for final round of refinement. For the clamp-loader:clamp complex, 18,727 movies were collected and 455,041 particles were sufficient for final round of refinement.

Data exclusions

Individual mutants that underwent bacteriophage selection which contained poor input counts (<10) are excluded from analysis. The standard procedure of cryo-EM data processing involves sorting of particles that are damaged or false-picked, which are unlikely to be refined correctly. This is implemented in cryoSPARC v4. All such exclusions are described in the Methods section.

Replication

To assess reproducibility of fitness effects resulting from deep mutagenesis, phage selection experiments were performed in biological triplicate (transformed *E. coli* culture split into three independent phage infections) within two independent trials (unique transformations of *E. coli* with saturation mutant libraries). For positions of interest located from the large library experiments, smaller libraries of individual positions (library size of 32 variants) was constructed and subject to bacteriophage selection. All replication attempts were successful. This results in a larger input variant sample size for critical analysis of fitness effects seen in the larger library. For cryo-EM, structures were determined from independent half datasets, which were compared to determine the resolution of the reconstruction.

Randomization

Division of datasets into two random halves was standard procedure for cryo-EM data processing. This is implemented in cryoSPARC v4.

Blinding

Blinding was not applicable to this study. No populations were preassigned to groups.

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☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging